

Dark-side effects:

A Revisit of Indian Central Public Sector Enterprises (CPSEs)

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Abstract

The study examines the congruity between capital structure and ownership structure on Indian Central Public Sector Enterprises (CPSEs). The econometric models are applied using both the full model (single equation) and reduced model (reduced-form equations) respectively. A balanced panel of 43 CPSEs with 663 firm-year observations have been analyzed for a period of sixteen years (2001-2016) respectively to examine the twin-effects of ownership structure and capital structure. The findings posit ownership indicators influence capital structure and their impact is very minimal. This resembles in the long-run firms are able to decompose their immediate effects of ownership. Moreover, in the long-run firms are unable to cope to the industry practices, nuances of stiff competition, managerial incompetence, and political influences asunder the grey-earnings of the CPSEs. Through empirical results, it is evident ownership structure revealed statistically negative results in all four measures such as *promoter concentration* (shares); *non-promoter concentration* (shares); *promoter shareholding* (%); and *non-promoter shareholding* (%) taking into consideration the inclusion of capital structure regressors. In addition, firm-specific regressors such as *enterprise value*, *material costs*, *investment turnover*, *firm indebtedness*, *asset tangibility*; *liquidity*; *creditors turnover*; *return on capital employed*; *cash flows*; *quality of income ratio*; and *firm size* have commendable effects on both leverage and ownership structure pertinently. Moreover, laying insights on State-Owned Enterprises (SOEs) along with CPSEs can enrich substantial evidence of both leverage and ownership structure rudiments. The study insights to support policy makers, regulatory bodies, academicians, corporate houses, and public investors to effectively decide the pattern of investment decisions based on a comprehensive outlook of both ownership structure and capital structure. A sound corporate governance framework with due transparency and accountability can usher the economic performance of CPSEs and thereby reduces agency costs. The study adds to the existing body of literature and provides a conclusive outlook of CPSEs with the support of panel data models.

Keywords: Capital Structure, Ownership Structure, Disinvestment, Privatisation, CPSEs.

Introduction

The term privatization slayed by the ambiguities of economic shocks, reprimand: One of the more complex issues in this area involves the interrelated questions of when to privatize, whether to privatize rapidly or slowly, what order to follow in privatizing firms (sequencing), whether to sell a State-Owned Enterprise (SOE) at once or in stages (staging), whether to restructure a SOE prior to sale (or to just restructure the SOE),.... , the complexity of the issue has limited the empirical work in this area. (p. 341) (Megginson & Netter, 2001). Taken literally, the study aims to provide interesting insights on the nature, functioning, progress, and the lacunas that hinder the public enterprises in the Indian context. The progress of Indian Central Public Sector Enterprises (hereafter, CPSEs) asunder the aftermath of Asian financial crisis. Poignantly, suffering from the internal resources constraints, refusal to attract foreign exchange reserves, inability to pay the principal and interest obligations on foreign borrowings haunted the economic growth meticulously. Besides the balance of payments crisis 1991 and the abolition of license-raj pact, India witnessed a major transformation in its economic policy to accept the structural reforms for its development. The new economic policy 1991 fine-grained liberalization, privatization, and globalization (LPG) that aimed at delicensing; free entry and exit for private players; divestment of government shareholding; listing arrangements on the stock exchanges; greater autonomy to profitable public-sector enterprises; and to rejuvenate loss-making enterprises by setting the Board for Industrial and Financial Reconstruction (BIFR). Broadly reflecting, performance contracts through Memorandum of Understandings (MoUs) were fortified to improve the performance of the public-sector undertakings (Chibber & Gupta, 2017). In contrast to replication, India abided to the structural adjustment programme adjudged by International Monetary Fund (IMF) for external borrowings in order to curtail losses borne by the Public Enterprises and to meet their budgetary requirements. Resultantly, CPSEs had to cope up with excessive bureaucratic checks; political interference in its effective functioning; lacunas of effective corporate governance; inconclusiveness in the capital structure decisions; managerial incompetence; public accountability; and finally, stock listing arrangements revamp their survival. Given these circumstances, the study prompts to revisit the spheres of Central Public Sector Enterprises (CPSEs) on the national front with respect to ownership concentration, operating and financial performance amidst the confluence of State-Owned Enterprises (SOEs) as a subset to usher economic growth and development.

The Dark-Side Effects: Disinvestment Dilemma

India, a transition economy had to overcome large fiscal deficit; lukewarm agricultural produce; inflationary pressures; inefficiencies in utilization of economic resources; heavy foreign debt have resented the credit rating; and the growth prospects needed a re-visit. As (Ghosh, 1992) elicited the economic reform policies extended stabilization, restructuring and globalization of the Indian economy to curb fiscal deficits and streamline public enterprises. The 1991 economic reforms have embellished a tremendous effect on the Indian economic scenario and prospects. More importantly, rupee was devalued by 18%-19% in quick stages; trade policy norms were curtailed; grants of new industrial licences; introduction of Monopolies and Restrictive Trade Practices Act; liberalization; attract Foreign Direct Investment; Privatization; financial sector reforms; and inflationary pressures posed a decisive effect on the Indian market. As highlighted by (Gollakota & Gupta, 2006), with the ebbing of economic pressures, India has evolved its business ownership consort from family ownership phase (Pre-Independence); public enterprises and socialist outlook phase (The License Raj pact); evolvement towards professional ownership and phase of social justice (Knowledge professionalism); and finally the era of foreign ownership (Liberalization). Indian CPSEs are classified under Maharatnas, Navaratnas and Miniratnas elusively. Maharatnas are listed enterprises in Stock exchange; with average annual turnover (>Rs. 25,000 crs.); average annual networth (> Rs. 15,000 crs.); and average annual net profit after tax (> Rs. 5,000 crs.) during the past 3 years; with significant presence worldwide; and enjoy Navratna status correspondingly. Similarly, Navratnas are Miniratna category – I with very good ratings under the MoU system over the past 3-5 years and report a composite score of above 60 in the selected parameters such as net profit/ net worth; manpower cost/ total production costs; PBDIT/ capital employed; PBIT/ turnover; EPS; and inter-sectoral performance respectively. Finally, Miniratnas are enterprises that have exhibited profits consecutively and have positive networth over the past 3 years are conferred Miniratna status. By way of a brief history during nineties, the then Union Government in force opted to launch the Divestment approach to attract capital from the stakeholders extensively. On the current reckoning, disinvestment diagnosis fetched revenues, in congruity to combat inflation and also aimed towards curbing the financial deficits. To a large part, strategic disinvestment and partial privatization improved the performance of government-owned enterprises. At this outset, the disinvestment proceeds were facilitated through the National Investment Fund to calibrate economic gains. Toward this purpose, the Department of Disinvestment was set up

in 1999 under Arun Shourie, who connoted “*these are the not the crown jewels (Ratnas) of India’s economy but bleeding ulcers*”. In this turbulent setting, privatization ostensibly was remarked as “strategic disinvestment” encouraging investment avenues to the stakeholders. For instance, (Gupta, Jain, & Yadav, 2011) argues disinvestment (sale of minority stakes) reported a positive impact on the performance of public sector undertakings as new promoters inject enough liquidity to ordain commercial value thereby improving efficiency, productivity and survival.

As (Palepu, *et al.*, 2000) connote the autism activities of the government should streamline both the work-force and the assets-in-place giving due emphasis on employee interests. In this respect, the committee reported the insights on centralized supervision that adhered transparency through decentralized implementation and control. Further, they augment the facets of privatization seeks to reap the fruits of infrastructural competitiveness, regulatory control based on operational efficiency, performance, and capital intensity. (Sengupta, 2002) recommended the performance evaluation of CPSEs through commercial viability and the total performance criterion (i.e. MoUs). Generally, MoU acts as an instrument that ensemble the performance of CPSEs based on social and financial objectives through autonomy as a prime transparent phenomenon (Plasman, 2008). Moreover, it reprimands a non-binding agreement between both the administrative ministry and the public enterprise by laying the mutual responsibilities and obligations to be performed. For example, (Gupta, 2005) reported disinvestment acts as a vital channel for privatisation whereby profit making CPSEs can opt for open market sales and likewise, loss-making CPSEs resort to strategic sales chronically. Moreover, the degree of partial privatisation has improved productivity such that optimal utilization of economic resources are put to use effectively. Said differently, (Selvam, 2007) enunciated disinvestment phenomenon impeded contrasting repurcussions for both developed (political entanglement) and developing economies (economic cum political entanglements) whereby questions the success of privatisation. In case of India, structural adjustment mechanism and the economic reforms though pro-active have lacked the upsurge to augment economic growth as multi-party governments churn power quite elusively. With the ebbing of economic pressures, (Nagaraj, 2005) elicited the need for institutional mechanism that distinguishes ownership and control of CPSEs as the disinvestment commission had to solve the modalities by reducing political and bureaucratic influences to meet the financing and investment targets. As (Naik, 2001) and (Sangeetha, 2005) concentrate on the economic reforms against financial upheavals through division of authority and co-operation; signing of

MoUs; modernization; delicensing; encourage private players; autonomy to managers; downsizing; budgetary support; and rehabilitation of sick CPSEs. Quite truly, (Arun & Nixon, 2000) proffered disinvestment debacle has been sought as an impediment to minimize the central government borrowings at a price that is detrimental to the rationalization and restructuring costs of CPSEs. Eventually, under-pricing of equities, accountability and transparency syndrome with limited support of the stakeholders has remarked the mediocre-success of disinvestment.

Amidst these repurcussions, (Reddy, Rao, & Mishra, 1990) ordained 2*2 matrix in deciding the CPSEs divestment of shareholdings such as financial success and also marketing structure the prime factors to choose enterprises eligible for the divestment process. In addition, social obligations towards the society and the categorization of similar groups were also considered to decide the candidates for disinvestment. Further, financially competitive (profitable) were hived-off and non-financially competitive (delved by financial reforms) CPSEs were provided opportunities to become financially viable and thereafter given to the private sector. In this respect, (Nanjundappa, 2005) proffered restructuring CPSEs through the approach of disinvestment alongside the arsenal of national employment policy. Moreover, nascent plan outlays, mobilize economic resources, meet budgetary requirements and reiterate employment upheavals act as churning reform measures to curb fiscal deficit. Concomitantly, (Sethi, 2005) enunciated the effects of both privatisation and disinvestment on profit-makers (enjoy monopoly), loss-makers (opt for restructuring) and sick CPSEs (sought rehabilitation) whereby economic, social and political implications can usher the success of disinvestment phenomenon. Succintly, the profits reaped by the CPSEs are not duly spent on the welfare measures such as education, infrastructure, and health services. Amidst these ambiguities, (Majumdar, 2005) described four vital rudiments on the privatisation debacle in India whereby the commendable areas were transparency (separation of ownership and control) and competitiveness (infrastructure), likewise, the areas for improvement include mitigation (restructuring exercise) and commitment (economic growth) respectively. As noticed by (Sengupta, 2002) the efficacy of MoUs can be channelized in performance evaluation on the basis of both internal and external constraints they can delve correspondingly. Internal constraints corroborate over-manning; inadequacy of resources; weak internal control systems; and lack of proper motivation among the personnel. Likewise, external constraints portray appointment of bureaucrats, political interferences; transfers; and assessment of contracts. (Shivendhu, 2008) elicited privatization impedes cost-benefit tradeoffs consorted

by the politicians through examining 4,700 privatization transactions tested on 43 countries that provided substantial evidence on institutional quality having prominent effects on partial privatization. Further, the findings highlighted that higher levels of corruption collided with higher levels of privatization in rather competitive industries, heterogeneous preferences and habits of citizens, and efficiency of private sector pose unanimous decline in the fruits of privatization.

Corporatization vs. Privatization

Looking ahead, the perusal of strategic reforms through the blend of corporatization and privatization augment the symphony of government contracts along with public and corporate managers embellished the performance of state-owned enterprises in 12 developing countries (Shirley, 1999). More importantly, both privatization and corporatization align together to reap success or failure, albeit of political costs stylized to countervail reforms or to accept alternative ownership mechanisms. With the ebbing of pressures, governments have sought corporatization as an essential instrument of economic policy (Aivazian, Ge, & Qiu, 2005). Driven by expectations, corporatization rudiments both the operating and financial conundrum thereby foster improvements in profitability, efficiency, turnover and productivity (Clo, *et al.*, 2015). On the global front, they have triggered economic development; level of capital markets; corporate governance norms; legal protection; ownership structure; firm performance; eventually capital and profit planning. Strikingly, (Liang, Ren, & Sun, 2015) connote corporatization oozes greater autonomy and control to SOEs in turn, reduces the direct involvement of government led by political parties and the bureaucrats. To a large part, the degree of corporatization reduces governmental intervention and interference with well-defined corporate policies. Succintly, non-commercial goals in lieu of costs or revenues envisage the performance parameters of SOEs (Bozec & Breton, 2003).

State-Owned Enterprises (SOEs)

In line with common wisdom, the International Monetary Fund (IMF) defines SOEs as “commercial entities owned or controlled by the government” (Allen & Alves, 2016). In fact, SOEs have transformed their business operations from non-profit to profit-seeking firms. Similarly, Organisation for Economic Co-operation and Development (OECD) defines “enterprises where the state has significant control, through full, majority or significant minority ownership” (OECD, 2005, p.11). Recently, the (Pwc, 2015) propound SOEs as companies whose ownership is fully controlled by the state, and which acts as the sole owner

or the owner of majority or material minority stake (authorized capital shares). Further, the material minority stake represents voting shares of at least 10%. As (Cazurra, *et al.*, 2014) noted the rationale behind SOEs from the precepts of both economic (curb market anomalies) and political ideologies sought by government bodies with respect to private ownership and control of the productive assets. On a more positive note, the latter ponders around the political strategies and ideology such as communism, nationalism, social and strategic. To be concrete, the communist ideology aimed at accretion of wealth under the control of private owners against the expense expunged on the workers. Succintly, in a nationalist view, the prime task of a government is envisaged to create SOEs in order to strengthen and acquaint the growth and development of the nation. Ultimately, the socialist ideology delved around the government's earnest efforts to invest elusively in the SOEs in order to fulfill the social objectives that are fostered with respect to education, poverty alleviation, health care, and infrastructure. Of course, the strategist ideology is abetted towards the creation and functioning of SOEs in congruity with the strategic landscapes that include defense and allied services. Under the social view, SOEs avoid illness and resort to cure market failures through facilitating effective pricing policies by considering both the social costs and production benefits (Shapiro & Willig, 1990). Similarly, the agency view contradicts managers of SOEs lack proper monitoring and the resurgence of political interference results in investment lacunas, excessive employment, stringent managerial incentives, and finally, pitiable product and location choices [(Vickers & Yarrow, 1988); (Shleifer & Vishny, 1994)]. (Luqmani & Quraeshi, 2011) reported plan and implementation model of privatizing SOEs for developing countries. In this respect, SOEs that are poor performers could be delineated to private entities whereby market inefficiencies are curtailed; stimulates economic growth and development; minimizes government deficits; cater public service demands; and finally offer better service amenities. As fostered by (Desvaux, Wang, & Zu, 2004) SOEs are baffled with incompetent bureaucrats and inefficient managers whereby the spawn of losses suffered by these enterprises are afloated through government subsidies, grants, and rehabilitation measures to churn their survival. Moreover, loss-making enterprises consort the need for privatization wherein private entities take the lead role to negate these enterprises from economic ills.

Ownership: A double-edged sword

Transition economies are most often identified with weak corporate governance standards that ensemble to curtail the inward Foreign Direct Investment (FDI) flows inclined with host

country benefits (Globerman & Shapiro, 2009). Nonetheless, the philosophy of SOEs are put forth under the presence of market failures, countervailing nuances of regulatory compliances and their inefficiency (Hart, Shleifer, & Vishny, 1997). Furthermore, the (Blume, 2013) elicit the ownership function of the state that includes the centralized model (one government body vested with the government's stake); decentralized model (governed by respective ministries); and the dual model (both a sector ministry and a separate central ministry are united). Interestingly, the existence of SOEs are linked to surpass market failures (i.e. industrialization, national security, employment and inequality) and regulatory upheavels as these problems are faced by economies with weak institutional environment. On both these occasions, namely the “*public commitment problem*” (unable to refrain from changing the regulatory regime) (Bortolotti & Perotti, 2007) and the “*private commitment problem*” (regulators unable to exercise effective control) (Hart, Shleifer, & Vishny, 1997) foresee the amicable repurcussions. This means that the functioning of SOEs are influenced to benefit self-interested politicians who sought their private gains over social welfare [(Shleifer & Vishny, 1993). SOEs are harnessed to achieve the social, political, and commercial objectives. This sluggishness happens when state ownership becomes passive whereby managers are not fully aware of their social goals; on the contrary, has to refrain from fine-grained political interference (Shleifer & Vishny, 1994). With proliferation, SOEs most often protect themselves from the threat of takeovers and bankruptcies with respect to policing management because, the state-government takes full control of the enterprise and also, access to state loans obtained from state-owned banks reduces the pitfalls of bankruptcy as these financial losses can be subsidized from the government finance. In this respect, the banks fail to conclusively monitor the SOEs commercial activities and their sustainability. Counter intuitively, this minimizes the need to contain costs and creates “*soft budget constraint*” whereby the transparency of SOEs are under nutshell if they are yet to have listing arrangements on a stock exchange (Globerman & Shapiro, 2009).

For example, (Arocena & Oliveros, 2012) compares both the pre and post privatization efficiency levels of Spanish SOEs using Data Envelopment Analysis (DEA) and found newly privatized firms increased substantially compared to their private counterparts. Concomitantly, the results reported private ownership far superior over SOEs due to the effects of agency-imperative between the owners and the managers (higher internal efficiency); the lethargic threats of mere closure; takeover and bankruptcy (personal emoluments under stake due to non-performance); and finally, the breadcrumbs of soft

budget constraints are beyond reach (sustain income and expenditure gaps) with higher autonomy and control vested in the hands of private enterprises. In a similar vein, (Huang, *et al.*, 2017) delves state ownership linked with higher dependence on SOEs pinpoint higher financial support from the home government, ordains the global outreach to expansion and the lack-lustre government intervention envisage the dark-side effects of SOEs.

In a nutshell, Japan dwindle around a decentralized framework for SOEs popularly known as Special Companies are dually monitored by the concerned ministry (pre-dominance of legislature) resembling a close encounter with the government in force. In the presence of economic shocks, Japan has its economic growth channelled around the philosophy of developmental state. In what follows, developmental state intended “*a state that puts economic development as the top priority of governmental policy and is able to design effective instruments to promote such a goal*” (Bagchi, 2000, p. 398). As (Meyer, *et al.*, 2009: 37) articulate that, “*While some domestic firms have also developed good corporate governance, the underdeveloped legal system has allowed local managers (or block shareholders) in many privatized firms to maximize their own benefits at the expense of corporate performance and hence welfare of (other) shareholders as well as stakeholders such as workers and government treasury. This is likely to account for the limited positive performance effects of privatization to domestic private owners as compared to the performance of firms privatized to foreign investors.*”

Confluence of Central Public Sector Enterprises vs. State-Owned Enterprises

Said differently, once noted by *Bureaucrats in Business* (World Bank, 1995) study on the reforms of SOEs in nine developing economies on the basis of region, size, economic conditions and income levels acutely confines political ambit as one of the prime reasons to augment the steady flow of privatization rather undermines the effective responsibility of economic and institutional restraints. Taken collectively, its opening words state: Bureaucrats are still in business. Despite more than a decade of divestiture efforts and the growing consensus that governments perform less well than the private sector in a host of activities, state-owned enterprises account for nearly as large a share of developing economies today as 20 years ago. Indeed, as data compiled for this study show, the size of the state-owned enterprise sector has significantly diminished only in the former socialist economies and a few middle-income countries. In most developing countries, particularly the poorest, bureaucrats run as large a share of the economy as ever. (p. 1). Moreover, SOEs reforms are aimed at privatization; financial sector reforms; competition policies; relationship between

both managers of SOEs; and the government. The key to success was based on profitability, savings-investment gap, and profitability of the SOEs.

Data, Variables and Hypotheses

Our study examines the congruity between capital structure & ownership structure using Indian non-financial CPSEs. The sample data for the study were extracted from Centre for Monitoring Indian Economy (CMIE) PROWESS IQ database for the period 2001 to 2016 respectively. CPSEs that are listed in the BSE/ NSE which are fully owned and controlled by the government are considered for the empirical analysis. STATA statistical package was used to conduct the empirical analysis. Moreover, enterprises that had incomplete financial data were excluded and the process led to a balanced panel of 43 CPSEs with 663 firm-year observations for a period of sixteen years correspondingly. Further, Banks, NBFCs, and other financial entities were excluded as their financial practice are different compared to the non-financial entities. Both the capital structure and ownership structure proxies, along with the control variables are reported in the Summary Statistics (see Table 1).

Methodology:

As (Al-Najjar & Taylor, 2008) connote both capital and ownership structure share similar set of causal variables and the firm-specific factors of ownership structure are extended subsets of the capital structure determinants. Further, augment that both these structure exemplify joint effects rather through simultaneous effects whereby there are not inter-dependent. To be concrete the paper aims to evident the congruity using both full and reduced-form models for capital and ownership structure. Inevitably, irrespective of the confluence of ownership structure on leverage, the latter seems to be rather under-identified (Al-Najjar & Taylor, 2008).

Similar to the equations followed by (Al-Najjar & Taylor, 2008), we use quite elusively on both capital structure and ownership structure respectively. The regression equations are given in the form:

$$Y_{it} = \alpha + \beta' X_{it} + \varepsilon_{it} \text{ (Pooled model)}$$

$$Y_{it} = \alpha_i + \beta' X_{it} + \varepsilon_{it} \text{ (Fixed effects model)}$$

$$Y_{it} = \alpha + \beta' X_{it} + (\varepsilon_{it} + \mu_i) \text{ (Random effects model)}$$

In this respect, the joint determinants affecting both capital and ownership structure are examined by using reduced-form models are provided in Tables 5 and 6 respectively: The reduced-form equations are confined as:

$$\begin{aligned} \text{stdr} [\text{ltdr}, \text{tdr}] = & \Pi_0 + \Pi_1 \text{mc_ev} + \Pi_2 \text{mcr} + \Pi_3 \text{pti} + \Pi_4 \text{invtr} + \Pi_5 \text{fa} + \Pi_6 \text{indebt} + \Pi_7 \text{fatr} + \\ & \Pi_8 \text{mfer} + \Pi_9 \text{ccl} + \Pi_{10} \text{ctr} + \Pi_{11} \text{roce} + \Pi_{12} \text{cf_rev} + \Pi_{13} \text{qir} + \Pi_{14} \text{tang} + \Pi_{15} \\ & \text{SZ} + \varepsilon_{it} \end{aligned}$$

$$\begin{aligned} \ln_ip_sh [\ln_np_sh, ip_p, np_p] = & \gamma_0 + \gamma_1 \text{mc_ev} + \gamma_2 \text{mcr} + \gamma_3 \text{pti} + \gamma_4 \text{invtr} + \gamma_5 \text{fa} + \\ & \gamma_6 \text{indebt} + \gamma_7 \text{fatr} + \gamma_8 \text{mfer} + \gamma_9 \text{ccl} + \gamma_{10} \text{ctr} + \gamma_{11} \text{roce} + \\ & \gamma_{12} \text{cf_rev} + \gamma_{13} \text{qir} + \gamma_{14} \text{tang} + \gamma_{15} \text{SZ} + v_{it} \end{aligned}$$

Theoretical Framework and Hypotheses Development

Tangibility [*tang*] and Leverage

Thies and Klock (1992) state that according to principle of asset maturity, net capital assets change the orientation of financing of short-term debt to long-term debt while stocks change equity to short-term and long-term debt. The empirical evidence reports a significant and positive relationship between tangibility and total debt (Rajan and Zingales, 1995; Titman and Wessels, 1988). Concomitantly, other studies found a positive relationship between tangibility and long-term debt, and a negative relationship between tangibility and short-term debt (Chittenden *et al.*, 1996; Stohs and Mauer, 1996; Van der Wijst and Thurik, 1993). In addition, few studies reported negative relationship between the two variables (Acaravci, 2004; Sayilgan *et al.*, 2006). In contrast, under the pecking order theory, firms with more tangible assets are less sensitive to information asymmetry problems and agency cost issues. Thus, they are more willing to issue equity than debt (Harris and Raviv, 1991). Titman and Wessels (1988) found that firms with fewer tangible assets may target higher debt ratio in order to stop managers from consuming more than optimum level of perquisites. This implies the negative relation between tangibility and leverage in accordance with pecking order theory.

H₁: There exists a negative relationship between tangibility and leverage.

Fixed Assets turnover ratio [*fatr*] and Leverage

The trade-off theory suggests a positive relationship between the share of fixed assets and debt ratio, since fixed assets serve as collateral for debt financing. In other words, firm will

obtain debt more easily as collateral value of fixed assets rise (Harris and Raviv, 1991; Myers, 1977; Myers and Majluf, 1984; Thornhill et al., 2004; Williamson, 1988). In the pecking order theory, however, firms that own more fixed assets have less asymmetrical information. Therefore, they tend to depend on equity financing. When we consider maturity, the pecking order theory suggests that share of fixed assets is positively related to long-term debt financing and negatively to the short-term debt financing (Feikadis and Rovolis, 2007; Qian et al., 2007). Most of the studies found a positive relationship between the share of fixed assets and leverage ratio (Booth et al., 2001; Chen, 2004; Fattouh et al., 2003; Pandey, 2004; Tang and Jang, 2007; Upneja and Dalbor, 2001; Zou and Xiao, 2006).

H₂: There exists a positive relationship between fixed assets turnover ratio and leverage.

Firm size [sz] and Leverage

The trade-off theory asserts a positive relationship between firm size and debt ratio because larger firms are better diversified and have a lower probability of experiencing financial distress. Lower bankruptcy costs allow large firms take advantage of leverage (Ang, 1992; Antoniou et al., 2002; Bevan and Danbolt, 2002; Homaifar et al., 1994; Wiwattanakantang, 1999). Empirical studies generally found a positive relationship between the two variables, and hence support for the trade-off theory (Lasfer, 1995; Dalbor and Upneja, 2002; Gaud et al., 2005; Huang and Song, 2006; Pandey, 2004; Qian et al., 2007; Sayilgan et al., 2006). In this respect, Rajan and Zingales (1995) empirically test firms' capital structure in G-7 countries and found that large firms borrow more as they are more diversified and have lower bankruptcy costs. Hence, there exists a positive association between firm size and leverage. Big companies are probably the most diversified and less likely to suffer liquidation. These firms have low direct costs when they decide to issue debt or equity. Also, they expect to use higher debt levels than small companies. Small businesses can have low levels of long-term debt and high levels of short-term debt due to conflict that can appear between shareholders and creditors (Michaelas et al., 1999; Titman and Wessels, 1988). On the contrary, the pecking order theory implies a negative relationship between firm size and debt ratio, since information asymmetrical is less severe issue in big firms. Thus, big firms' cost of capital should be less than that of small firms (Rajan and Zingales, 1995; Zou and Xiao, 2006). Further, Frank and Goyal (2008) noted that large firms face lower adverse selection costs which enable them to easily issue equity.

H₃: There exists a positive relationship between firm size and leverage.

Profitability [*roce*] and Leverage

When firms are highly profitable, they will use these profits for their future growth and develop investment opportunities and therefore, they avoid using debt and issuing equity. Conversely, less profitable firms but with growth prospects will use debt to finance their investment projects since they do not have own internal funds. This in fact shows the negative relation between *roce* and leverage. Accordingly, debt ratio falls as profit increases for the firms financing themselves with debt (Myers and Majluf, 1984). Similarly, Allen (1993) argues that debt financing are less prone to information asymmetry problem than equity and more often, are likely to be undervalued in the market. The pecking order and trade-off theories also have opposite implications about the relationship between profitability and debt ratio. The pecking order theory opts for a negative, while trade-off theory opts for a positive relationship between the two variables (Benito, 2003; Krasker, 1986; Myers, 1984; Myers and Majluf, 1984; Narayanan, 1988; Qian et al., 2007). According to the trade-off theory, high profitability level renders a high level of borrowing capacity. Thus, the trade-off theory hypothesizes a positive relationship between profitability and debt level (Frank and Goyal, 2003; Um, 2001). As profit increases, firms issue more debt to shield the profits from taxation. Consequently, with higher profits generated, higher debt ratio is expected (Jensen, 1986). Graham (2000) by estimating the cost and benefit of debt, found that larger and the more profitable firms with low financial distress exposure are expected to use more debt, implying the positive relation between profit and leverage in support of trade-off theory. According to the pecking order theory, high-profit firms outperform low-profit firms in terms of using retained earnings in internal financing. As a result, the pecking order theory suggests a negative relationship between profitability and debt level (Myers, 1984; Myers and Majluf, 1984). Nevertheless, empirical studies generally found a negative relationship (Acaravci, 2004; Allen, 1991; Barton and Gordon, 1988; Chen, 2004; Huang and Song, 2006; Pandey, 2004; Sayilgan et al., 2006; Tong and Green, 2005; Wiwattanakantang, 1999).

H4: There exists a positive relationship between profitability and leverage.

Free cash flows [*cf_rev*] and Leverage

The trade-off theory suggests a positive relationship between free cash flows and debt ratio, since firms owning big amount of free cash flows are exposed to less risk and borrow more easily in capital market (Benito, 2003; Jensen, 1986; Stulz, 1990). However, a negative relationship is implied by the pecking order theory, since it requires an increase in internal funds arising from free cash flows. These incremental cash flows would be regarded as a

financing source (Myers, 1984; Myers and Majluf, 1984). Empirical studies obtained conflicting evidence on the relationship between free cash flows and leverage ratio (Colombo, 2001; Jensen, 1986; Stulz, 1990; Westphalen, 2002).

H₅: There exists a positive relationship between free cash flows and leverage.

Liquidity [*ccl*] and Leverage

POT states liquidity and leverage are negatively related (Titman and Wessels, 1988) whereas TOT exerts that firms choose to have high debt levels in order to enjoy tax advantages and generates a positive relationship between liquidity and debt. Likewise, Jensen (1986) reveals that businesses rich in cash tend to obtain a new debt to avoid managers not using free cash flows efficiently confines a positive relationship with liquidity.

H₅: There exists a negative relationship between free cash flows and leverage.

Inventories and Leverage

As (Kruger, 2005) corroborates inventories ensemble a major portion of the firm's current assets. (Sprague & Wacker, 1996) opine firm's viability can be jeopardized due to mismanagement of inventories whereby its effective control is questionable. Subsequently, (Meyer, 1991) asunder *cost savings* phenomenon by incorporating best practices on inventory management mitigates productivity loss, time, and labour discrepancies.

H₅: There exists a positive relationship between free inventories and leverage.

Discussion & Interpretation

Observing both the full model and reduced-form model of ownership structure interesting results have been derived. The full model including capital structure regressors evident as follows: *ln_ip_sh* quite clearly is negatively related to *ltdr* and *tdr* in all four cases whereas *stdr* shows mixed results. *ln_np_sh* pertinently negative in most cases, except for pooled ols where both *ltdr* and *tdr* are positive. *ip_p* consecutively negative in all cases, whereas in case of *tdr* it is positive. *np_p* positive in case of *stdr* and *ltdr* for pooled ols where as negative in other cases; likewise, *tdr* is negative for *tdr* (pooled ols) whereas positive in all other cases.

In particular, for capital structure (both full and reduced-form models), mixed results were found. Succintly, for *stdr* mixed impact was found on *sz* (pooled); *ltdr* on *roce* (pooled), *cf_rev* (tfe); *tdr* on *fa* and *fatr* (pooled), *tang* (fe), *ccl* (tfe) respectively. Both observing both full and reduced models *mcr* and *invtr* showed positive results throughout; *pti* and *ctr* portrayed negative association likewise; *mc_ev*, *fa*, *fatr*, *mfer*, *ccl*, *roce*, *qir*, *tang* and *sz*

showed consistent results in most cases for *stdr*, *ltdr* and *tdr* respectively. Mixed results (comparing both full and reduced-form models) with alternative signs were found in cases of ownership structure measures. For example, *ln_ip_sh* shows mixed impact on *mc_ev* (fe, tfe and re); *ln_np_sh* on *qir* (pooled), *mc_ev* (fe & re), *mcr* (re), *ctr* (fe), *invtr* (tfe); *ip_p* on *fa* (fe), *mfer* (fe, tfe and re), *invtr* (tfe & re), *ccl* & *tang* (tfe), *sz* (re); and *np_p* on *invtr*, *fa*, *mfer* (fe); *mc_ev*, *indebt*, *ccl*, *tang* (tfe); *mfer* and *ccl* (re) respectively. From the ownership structure measures, *pti*, *fatr*, *ctr*, *roce*, *cf_rev* and *sz* show significant results in most cases; likewise, *mc_ev*, *mcr*, *invtr* and *mfer* show mixed results in most cases; and finally, *fa*, *indebt*, *ccl*, *qir* and *tang* show rather minor mixed effects. Compared to ownership structure, capital structure models provide more consistent results meticulously.

Table 1 and Table 2 portray the estimates for both the capital structure and ownership structure equations respectively. In both cases, all of the pertinent regressors including the joint determinants are given due importance. As mentioned earlier, three measures are employed to determine the effects of capital structure (*stdr*, *ltdr* and *tdr*). Similarly, four measures are used for ownership structure (*ln_ip_sh*, *ln_np_sh*, *ip_p* and *np_p*) correspondingly. In the former case (See, Table 1), ownership structure proxies are considered as independent regressors and in case of latter (See, Table 1), capital structure measures are used as independent variables respectively. Subsequently, three separate equations are employed for capital structure and four separate equations for ownership structure are reported in Table 1 and Table 2. With the given panel data, all the encompassing equations are examined using pooled ols, fixed effects, time fixed effects, and random effects model elusively. Further, time-invariant heterogeneity among the firms are controlled in case of the latter three techniques. To facilitate the estimated results statistically, we report the effect test (chow test) for time-fixed effects; Hausman test for fixed effects; and Lagrange multiplier test for random effects respectively. If the effect test is statistically significant, we conclude the need for year dummies in the time fixed effects model. In addition, if the hausman specification test is significant we prefer fixed effects model over random effects model. Also, if the Lagrange multiplier test is significant we prefer random effects model (i.e. presence of firm heterogeneity holds a decisive effect) over pooled ols respectively. Moreover, as from the full models of both capital and ownership structure; we also provide an insight on re-estimating the same models using reduced-form equations and their results are reported in Table 3 and Table 4 correspondingly.

Quite clearly from Table 1 to Tables 4, the effect test (chow test) shows significant results for capital structure (both full and reduced-form); whereas, for ownership structure except for *ln_np_sh* (both full and reduced-form) all other measures *ln_ip_sh*, *ip_p*, and *np_p* for both full and reduced-form model showed insignificant probability value greater than 0.05. Hence, there is no relevance of time-fixed effects in such cases. The Hausman specification test is statistically significant in all cases of capital structure both for full and reduced-form models respectively. Further, the Hausman specification test was statistically significant in all cases for capital structure model; whereas, for ownership structure *ln_ip_sh* (for full model) and *ip_p* (for both full and reduced-form model) evident negative co-efficients. Hence, it is not clear whether there are any fixed effects and said differently, random effects model are preferred over fixed effects in such cases.

Concomitantly, the Lagrange multiplier shows statistically significant results in all cases for the capital structure and ownership structure models. It can be inferred that the presence of random effects over pooled ols is clearly evident. We report there exist differences between the CPSEs in examining both the capital and ownership structure; and also, pertinent differences exist between the independent regressors used. We conclude that means of the aforesaid differences between CPSEs are normally distributed and hence, purely random. Likewise, Hausman test is statistically significant for *ln_np_sh* and *np_p* (full model) and for *ln_ip_sh*, *ln_np_sh* and *np_p* (reduced-form model). To connote, the firm-specific determinants that conclude differences between *ln_np_sh* and *np_p* (full model) and for *ln_ip_sh*, *ln_np_sh* and *np_p* (reduced-form model) are unable to be examined randomly.

Through a clear investigation of both the full model and the reduced-form model, it is conclusive from the capital structure equation (i.e. *stdr*, *ltdr*, and *tdr*) that the joint determinants such as *mc_ev*, *mcr*, *pti*, *invtr*, *fa*, *indebt*, *fatr*, *mfer*, *ccl*, *ctr*, *roce*, *cf_rev*, *qir*, *tang*, and *sz* confine similar results in both the cases respectively. Interestingly, ownership structure equation (i.e. *ln_ip_sh*, *ln_np_sh*, *ip_p*, and *np_p*) portray slight variations in their relationships. Comparing both the full and reduced-form models, the latter reported that *mc_ev* [i.e. *ln_ip_sh* (+ve), *np_p* (-ve)]; *mfer* [i.e. *ip_p* (-ve), *np_p* (+ve)]; *ctr* [*ln_ip_sh* (+ve)]; and *qir* [*ln_np_sh* (-ve)] reprimand mixed results elusively. Subsequently, other firm-specific regressors such as *mcr*, *pti*, *invtr*, *fa*, *indebt*, *fatr*, *ccl*, *roce*, *cf_rev*, *tang*, and *sz* are in congruity with the full model of ownership structure respectively.

The pooled OLS model for capital structure measures such as *stdr*, *ltdr*, and *tdr* for the full model reported r^2 of 70.8%, 75.2% and 71.6% respectively. Similarly, for the reduced form model the reported r^2 were 69.6%, 73.6% and 69.9% correspondingly. It can be inferred that with the influence of ownership indicators (*ln_ip_sh*; *ln_np_sh*; *ip_p*; and *np_p*) in the full model the increase in the r^2 is around 1.2%, 1.6% and 1.7% respectively. Quite clearly, ownership indicators influence capital structure and their impact is very minimal. Both fixed effects and random effects f-statistics reported significant values. In line with the time fixed effects for the full model, *ltdr* had a p-value of 0.0710 (i.e. $\text{prob}>F$ is >0.05) the null hypothesis (i.e. all years joint coefficients = 0) is accepted and thereby there is no need of time fixed effects in this case and vice-versa applies to *stdr* and *tdr*. The Hausman test for both full and reduced model is significant (i.e. $\text{Prob} > \chi^2$ is <0.05) and hence fixed effects model is the desired one. Further, random effects test using Breusch-Pagan Lagrange Multiplier test was performed to see the confluence whether to go for random effects or the pooled OLS. The null hypothesis in this case (i.e. variance across firms = 0). From the reported values of LM test, (i.e. the $\text{prob}>\chi^2$ is 0.0000) and hence, we accept random effects model over pooled OLS.

Likewise, for pooled OLS in case of ownership structure measures such as *ln_ip_sh*; *ln_np_sh*; *ip_p*; and *np_p* for the full model reported an r^2 of 75.3%, 70.5%, 47.2% and 44.2% respectively. Further, for the reduced form model the results portrayed 74.5%, 69.9%, 46.7% and 44.1% correspondingly. Interestingly, the influence of the regressors with the full model showed hardly 1% increase in the r^2 . Both in case of fixed effects (f-statistics) and random effects (Wald statistics) are significant. In line with the time fixed effects for the full model, *ln_ip_sh*, *ip_p* and *np_p* had a p-value of 0.1562, 0.6666, and 0.0501 and for the reduced form model 0.394, 0.8611, and 0.0829 (i.e. $\text{prob}>F$ is >0.05) the null hypothesis (i.e. all years joint coefficients = 0) is accepted and thereby there is no need of time fixed effects in this case and vice-versa applies to *ln_np_sh*. The Hausman test for both full and reduced model for *ln_np_sh* is significant (i.e. $\text{Prob} > \chi^2$ is <0.05) and hence fixed effects model is preferred. Whereas in case of full model, *ln_ip_sh* and for reduced model *ip_p*, the hausman test reports negative values (i.e. the data rather fails to meet the asymptotic assumptions) and vice-versa for the other ownership measures. Thus, it is inconclusive whether to go for random effects model as the reported χ^2 value is negative. The LM test for both the full and reduced model are statistically significant and hence we prefer random effects over POLS.

From the full model of leverage, it can be inferred by pooled ols, *ln_ip_sh* has a negative impact for *stdr* (-2.06), *ltdr* (-4.12) and *tdr* (-6.92). In case of *ip_p*, *ln_np_sh*, *np_p* are negative for *stdr* and positive for *ltdr* and *tdr*. This resembles in the long-run firms are able to decompose their immediate effects of ownership. In case of *fa*, *indebt*, and *fatr* for *stdr* there is a significant positive association whereas for *ltdr* and *tdr* there is a negative relation. Quite clearly, in the long-run firms are unable to cope to the industry practices, nuances of stiff competition, managerial incompetence, and political influences asunder the CPSEs. In case of *ctr* and *roce* there exists a negative relationship for all leverage measures. As most of the stakes lie in the armoury of the central government, the *roce* is negative raises a question as to the performance of CPSEs is tact or not. *tang* has a negative relationship for *stdr* (-3.36), *ltdr* (-2.81), and *tdr* (5.71); *size* has a positive relationship with 1.80, 1.32, and 2.79; *qir* has a positive relation for *stdr* (1.32) whereas for *ltdr* (-3.02) and *tdr* (-1.56) it is negative; *cf_rev* and *mcr* has a positive relationship in all cases whereby the material costs borne by these firms are easily met; *ccl* proclaims a mixed result with negative relationship for *stdr* (5.61) and *tdr* (-4.15) whereas for *ltdr* (2.02) a positive relation exists. *mfer* has negative relation for *stdr* (-7.02) that ensembles firms are unable to meet their operating costs; whereas in case of *ltdr* (41.00) and *tdr* (23.8) there is a positive relationship whereby firms are able to meet their operating expenses quite easily. In case of *mc_ev*, for *stdr* (4.58) there is a positive effect whereas for *ltdr* (-18.7) and *tdr* (-16.00) proclaims is worrisome as these firms are unable to attract investors to go for their shares.

In case of reduced model for leverage, *pti* and *ctr* showed consecutively negative relationship with leverage; likewise, *mcr* and *invtr* showed positive association; *mc_ev*, *fa* and *indebt* showed positive association with *stdr* for pooled ols, *fe*, *tfe* and *re*; whereas for *ltdr* and *tdr* they showed negative relationship; *cf_rev* showed positive results for *stdr*, *ltdr* and *tdr* in case of pooled ols, *fe*, and *re* whereas for *tfe* it showed negative association with *ltdr* and *tdr*; *tang* is negatively associated in case of *stdr* and *tdr* in all cases except for *ltdr* it is positive; *sz* for *ltdr* and *tdr* are positively related whereas it shows mixed results for *stdr* in case of pooled ols and *tfe* it is negative; similarly it is quite opposite in case of *qir* as compared to *sz*; *roce* is negative in all cases for *stdr* whereas for *ltdr* and *tdr* it is positively related; *fatr* is positive in all cases for *stdr* and *tdr* whereas for *ltdr* it is negative; *mfer* is negative in all cases for *stdr* whereas it is positive for *ltdr* and *tdr* in all cases; *ccl* is negative in all cases for *stdr* whereas it shows mixed for both *ltdr* and *tdr* respectively.

In case of full model of ownership structure, *ln_ip_sh*; *ln_np_sh*; *ip_p*; and *np_p* were used as dependent variables. From the full model of ownership structure, it can be inferred by pooled ols *ln_ip_sh* and *ip_p* had a decisive negative impact on both *stdr* (-2.071) (-22.94) and *ltdr* (-0.417) (-7.324) whereas it showed a positive association against *tdr* (0.279) (14.29) respectively. *ln_np_sh*, *np_p* are negative for *stdr* and positive for *ltdr* and *tdr*. This resembles in the long-run firms are able to decompose their immediate effects of ownership. In case of *fa*, *indebt*, and *fatr* for *stdr* there is a significant positive association whereas for *ltdr* and *tdr* there is a negative relation. Quite clearly, in the long-run firms are unable to cope to the industry practices, stiff competition, managerial incompetence, political influences asunder the CPSEs. In case of *ctr* and *roce* there exists a negative relationship for all leverage measures. As most of the stakes lie in the armoury of the central government, the *roce* is negative raises a question as to the performance of CPSEs is tact or not. *tang* has a negative relationship for *stdr* (-3.36), *ltdr* (-2.81), and *tdr* (5.71); *size* has a positive relationship with 1.80, 1.32, and 2.79; *qir* has a positive relation for *stdr* (1.32) whereas for *ltdr* (-3.02) and *tdr* (-1.56) it is negative; *cf_rev* and *mcr* has a positive relationship in all cases whereby the material costs borne by these firms are easily met; *ccl* proclaims a mixed result with negative relationship for *stdr* (5.61) and *tdr* (-4.15) whereas for *ltdr* (2.02) a positive relation exists. *mfer* has negative relation for *stdr* (-7.02) that ensembles firms are unable to meet their operating costs; whereas in case of *ltdr* (41.00) and *tdr* (23.8) there is a positive relationship whereby firms are able to meet their operating expenses quite easily. In case of *mc_ev*, for *stdr* (4.58) there is a positive effect whereas for *ltdr* (-18.7) and *tdr* (-16.00) proclaims is worrisome as these firms are unable to attract investors to go for their shares. *ln_ip_sh* is positive in case of *stdr* for *fe*, *tfe* and *re* whereas for *ltdr* (-5.09) (-4.36) (-4.09) and *tdr* (-4.57) (-4.16) (-4.97) it is negative. *ip_p* is positive in all cases for *tdr* whereas it shows mixed results for both *stdr* and *ltdr*. *np_p* is positive in all cases for *ltdr* whereas it shows mixed results for *stdr* and *ltdr*. *mc_ev* is positive in all cases for *stdr* whereas it is negative in all cases for *ltdr* and *tdr*. *mcr* is positive in all 4 cases such as *ols*, *fe*, *tfe*, and *re*. *pti* is negative in all cases. *invtr* is positive in all cases whereas its effects are quite minimal. *fa* and *indebt* is positive in case of *stdr* whereas for all the three remaining cases on *ltdr* and *tdr* it is negative. *fatr* is positive for both *stdr* and *tdr* whereas it is negative for *ltdr*. *mfer* is negative for *stdr* whereas for all other cases it is positive. *ccl* and *ctr* are negative for both *stdr* and *tdr* whereas it is positive for *ltdr*. *ctr* is negative in all cases. *roce* is negative in case of *stdr* where as for all other cases it is positive. *cf_rev* is positive in all cases whereas its effects are minimal. *qir*

is positive for *stdr* whereas for *ltdr* and *tdr* in all cases it is negative. *tang* is negative in all cases with significant effect on leverage. *sz* has significant effect on leverage for all cases.

In case of full model of ownership structure, *ln_ip_sh*; *ln_np_sh*; *ip_p*; and *np_p* were used as ownership measures. From the full model of ownership structure, it can be inferred by pooled ols *ln_ip_sh* and *ip_p* had a decisive negative impact on both *stdr* and *ltdr* whereas it showed a positive association against *tdr* respectively. *ln_ip_sh* for *stdr* was positive in fe, tfe, and re except in case of pooled ols (-2.071) showing a negative association whereas for *ln_np_sh* it was vice-versa for *ltdr*. *ln_ip_sh* in case of *tdr* was negative for fe, tfe, and re and for only *stdr* it was positive. *ln_np_sh* was negative for fe, tfe and re as *ltdr* was lower compared to *stdr* and eventually for pooled ols and fe it was positive for *tdr* and negative for tfe and re respectively. *ip_p* was negative for pooled ols in case of *stdr* and *ltdr* whereas it was positive for *tdr*; in case of fe, tfe and re it was positive for *stdr* and negative for *ltdr* and *tdr*. *np_p* was positive for both *stdr* and *ltdr* whereas it was negative for *tdr*; whereas it was negative for *stdr* and *ltdr* in case of fe, tfe and re and it was positive for *tdr* respectively. *ln_ip_sh* and *ln_np_sh* for *ltdr* and *tdr* were negative for fe, tfe and re and was positive in case of pooled ols for *tdr*. *ip_p* was negative for pooled ols for *stdr* and *ltdr* and in case of *np_p* vice-versa. *ip_p* was positive for *tdr* whereas *np_p* was negative for *tdr*; in case of fe, tfe and re *stdr* and *ltdr* were negative for *ip_p* and positive for *stdr* and *ltdr* in case of *np_p*. Both *ip_p* and *np_p* showing contrasting results whereby *ip_p* (negative) for pooled ols and *np_p* (positive) and also, *ip_p* (negative) for *stdr* and *ltdr* and *np_p* (positive) respectively.

The independent regressors for both full and reduced form model provide interesting insights: *ln_ip_sh* and *ln_np_sh* for the full model were negatively related whereas in case of reduced model they were positive for fe, tfe and re respectively. *mcr* in case of random effects for *ln_np_sh* for full model was positive and negative for reduced model; *invtr* for *ip_p* and *np_p* were negative for full model and positive for reduced model; *fa* for *ip_p* and *np_p* showed opposite results for both full model and reduced model for fixed effects. *indebt* in case of *np_p* for tfe was positive for full model and negative for reduced model; *fatr* was positive associated for the full model and negatively related for reduced model in case of pooled ols; *mfer* for *ip_p* for fe, tfe and re was positive for full model and negative for reduced model; *ccl* negatively related for full model and positive related for reduced model in case of tfe and re; *ctr* for *ln_np_sh* and *ln_ip_sh* for fe and tfe showed negative results for full

model and positive for reduced model with hardly any effect; *qir* in case of *ln_ip_sh* and *ln_np_sh* showed positive results for full model and negative results for reduced model in case of pooled ols; *tang* for *ip_p* and *np_p* showed opposite results in case of both full model and reduced model; *sz* showed positive impact on full model and negative relation in case of reduced model respectively.

Implications of the Study

The findings posit ownership indicators influence capital structure and their impact is very minimal. This resembles in the long-run firms are able to decompose their immediate effects of ownership. Moreover, in the long-run firms are unable to cope to the industry practices, nuances of stiff competition, managerial incompetence, and political influences asunder the grey-earnings of the CPSEs. Through empirical results, it is evident ownership structure revealed statistically negative results in all four measures such as *promoter concentration* (shares); *non-promoter concentration* (shares); *promoter shareholding* (%); and *non-promoter shareholding* (%) taking into consideration the inclusion of capital structure regressors. In addition, firm-specific regressors such as *enterprise value*, *material costs*, *investment turnover*, *firm indebtedness*, *asset tangibility*; *liquidity*; *creditors turnover*; *return on capital employed*; *cash flows*; *quality of income ratio*; and *firm size* have commendable effects on both leverage and ownership structure pertinently. Moreover, laying insights on State-Owned Enterprises (SOEs) along with CPSEs can enrich substantial evidence of both leverage and ownership structure rudiments. The study insights to support policy makers, regulatory bodies, academicians, corporate houses, and public investors to effectively decide the pattern of investment decisions based on a comprehensive outlook of both ownership structure and capital structure. A sound corporate governance framework with due transparency and accountability can usher the economic performance of CPSEs and thereby reduces agency costs. The theoretical implications of the study confine the grassroots of trade-off theory to be a better prospect for the CPSEs; while agency cost theory facilitates the ownership concentration debacle; and the insights of market timing theory needs to be given due importance respectively. Moreover, the attempt to unlock the hidden values of CPSEs becomes successful when the market obtains a transparent view of their functioning. Business factors such as growth in sales volume or growth in PAT must drive the firms towards fair pricing in the capital market and their ability to raise debt.

Table1: Summary Statistics

VARIABLES	N	mean	sd	min	max	var	skew.	kurt.	p25	p50	p75	p95
<i>stdr</i>	663	0.22	0.17	0.01	0.93	0.03	1.31	4.34	0.09	0.17	0.31	0.61
<i>ltdr</i>	663	0.23	0.21	0	0.95	0.04	1.13	3.70	0.06	0.18	0.34	0.72
<i>tdr</i>	663	0.44	0.25	0.02	1	0.06	0.31	2.16	0.22	0.41	0.62	0.91
<i>in_ip_sh</i>	663	19.50	1.89	13.51	23.09	3.58	-0.24	2.85	17.97	19.64	20.75	22.50
<i>in_np_sh</i>	663	18.43	1.72	12.72	21.70	2.97	-0.68	3.60	17.54	18.50	19.73	20.97
<i>ip_p</i>	663	69.65	20.08	14.04	99.59	403.1	-0.54	2.46	55.53	74	88.57	93.56
<i>np_p</i>	663	29.34	18.72	0.41	84.62	350.4	0.43	2.19	11.43	26	44.47	62.16
<i>mc_ev</i>	663	0.93	0.49	0	3.130	0.24	0.87	4.72	0.61	0.95	1.17	1.84
<i>mcr</i>	663	0.32	0.27	0	1.04	0.07	0.68	2.57	0.07	0.29	0.49	0.89
<i>pti</i>	663	16.28	15.40	0.07	87.63	237.1	1.66	6.76	4.08	12.04	24.59	46.37
<i>invttr</i>	663	3.54	5.54	0.03	49.74	30.79	4.12	26.67	0.70	1.51	4.18	12.50
<i>fa</i>	663	35.75	16.58	2	92	275.0	0.33	3.21	23	37	48	59
<i>indebt</i>	663	25.70	28.94	0.18	176.6	837.4	1.51	5.20	4.33	11.74	41.45	86.67
<i>fatr</i>	663	6.06	9.91	0.01	94.30	98.25	4.58	31.81	0.98	2.51	7.62	20.33
<i>mfer</i>	663	0.06	0.08	0	0.61	0.00	2.03	7.84	0.01	0.02	0.09	0.23
<i>ccl</i>	663	0.68	0.78	0	4.800	0.61	2.38	9.41	0.20	0.38	0.79	2.35
<i>ctr</i>	663	6.66	5.97	0.05	33.53	35.65	1.52	5.51	2.33	4.63	9.31	18.90
<i>roce</i>	663	4.48	5.11	-6.05	35.96	26.12	1.91	7.80	0.96	2.64	6.58	14.86
<i>cf_rev</i>	663	0.43	0.95	-0.13	9.81	0.90	6.41	54.16	0.02	0.16	0.49	1.47
<i>qir</i>	663	0.58	0.79	-1.69	4.82	0.63	0.92	9.42	0.34	0.63	0.85	1.61
<i>tang</i>	663	0.28	0.21	-1.15	0.86	0.04	-0.48	6.06	0.10	0.30	0.42	0.63
<i>sz</i>	663	11.38	1.64	7.03	14.78	2.69	-0.11	2.49	10.22	11.43	12.53	14.12

Table 2: Correlation Matrix

	<i>stdr</i>	<i>ltdr</i>	<i>tdr</i>	<i>ip_sh</i>	<i>np_sh</i>	<i>ip_p</i>	<i>np_p</i>	<i>mc_ev</i>	<i>mcr</i>	<i>pti</i>	<i>invtr</i>	<i>fa</i>	<i>indebt</i>	<i>fatr</i>	<i>mfer</i>	<i>ccl</i>	<i>ctr</i>	<i>roce</i>	<i>cf_rev</i>	<i>qir</i>	<i>tang</i>	<i>sz</i>
<i>stdr</i>	1																					
<i>ltdr</i>	-0.15	1																				
<i>tdr</i>	0.52	0.74	1																			
<i>ip_sh</i>	-0.37	0.21	-0.08	1																		
<i>np_sh</i>	-0.42	0.25	-0.05	0.73	1																	
<i>ip_p</i>	-0.1	-0.01	-0.09	0.64	0.03	1																
<i>np_p</i>	0.04	0.02	0.06	-0.56	0.05	-0.95	1															
<i>mc_ev</i>	0.1	-0.73	-0.57	-0.12	-0.23	0.17	-0.16	1														
<i>mcr</i>	0.26	0.27	0.41	-0.19	0.07	-0.39	0.33	-0.29	1													
<i>pti</i>	-0.52	0.01	-0.35	0.46	0.31	0.38	-0.37	0.1	-0.44	1												
<i>invtr</i>	0.64	0.15	0.49	-0.19	-0.32	-0.04	0	-0.1	0.19	-0.4	1											
<i>fa</i>	0.33	-0.21	0.05	-0.19	-0.3	0.2	-0.19	0.18	-0.14	-0.15	0.15	1										
<i>indebt</i>	-0.07	-0.63	-0.59	-0.1	-0.11	0.07	-0.04	0.6	-0.29	0.23	-0.24	0.13	1									
<i>fatr</i>	0.61	-0.24	0.2	-0.29	-0.39	0.01	-0.07	0.26	0.1	-0.27	0.46	0.21	0.1	1								
<i>mfer</i>	-0.18	0.28	0.09	-0.02	-0.02	-0.04	0.08	-0.17	-0.06	-0.08	-0.1	-0.21	-0.21	-0.18	1							
<i>ccl</i>	-0.32	-0.32	-0.51	0.05	-0.04	0.19	-0.18	0.49	-0.25	0.4	-0.23	0.01	0.48	-0.01	-0.02	1						
<i>ctr</i>	0.11	-0.09	0	-0.1	0.07	-0.29	0.33	-0.03	0.24	-0.33	0.29	-0.02	-0.07	0.06	0	0.01	1					
<i>roce</i>	0.02	-0.19	-0.14	-0.15	-0.12	0	0.01	0.21	-0.13	0.1	0.01	0.39	0.56	0.06	-0.29	0.21	0.01	1				
<i>cf_rev</i>	-0.16	0.19	0.04	0.16	0.11	0.14	-0.13	0	-0.12	0.38	-0.14	-0.08	-0.02	-0.1	0.01	0.21	-0.23	-0.01	1			
<i>qir</i>	0.17	-0.22	-0.07	-0.05	-0.07	-0.04	0.03	0.05	0	-0.1	0.13	-0.01	-0.03	0.13	-0.14	-0.02	0.03	-0.2	-0.18	1		
<i>tang</i>	-0.33	0.06	-0.15	0.16	0.3	-0.13	0.17	-0.25	-0.13	0	-0.21	-0.24	-0.17	-0.43	0.08	-0.09	0.1	-0.24	-0.2	0.25	1	
<i>sz</i>	-0.14	0.31	0.18	0.68	0.65	0.31	-0.26	-0.26	0.08	0.16	-0.06	0.08	-0.12	-0.23	-0.17	-0.09	0.05	0.21	0.1	-0.16	0	1

Notes: *stdr* is the short-term debt ratio measured as short-term debt divided by total assets; *ltdr* is the long-term debt ratio measured as long-term debt divided by total assets; *tdr* is the total-debt ratio measured as the sum of short-term debt and long-term debt divided by total assets; *in_ip_sh* is the total number of shares held by indian promoters calculated as the natural logarithm of the number of shares owned by Indian promoters; *in_np_sh* is the total number of shares held by non-promoters calculated as the natural logarithm of the number of shares owned by non promoters; *ip_p* is the proportion of indian promoters shareholding (%) calculated as the percentage of ownership stakes held by indian promoters; *np_p* is the proportion of non promoters shareholding (%) calculated as the percentage of ownership stakes held by non promoters; *mc_ev* is the market capitalization to enterprise ratio measured by market capitalization divided by enterprise value; *mcr* is the material cost ratio measured by raw materials, stores and spares divided by net sales; *pti* is the sales growth measured by the natural logarithm of total revenues; *invtr* is the investment turnover ratio measured by net sales divided by net worth; *fa* is the age of the firm measured by the number of years since the date of inception; *indebt* is the firm indebtedness measured by profit before depreciation, interest, taxes and amortization divided by total interest expenses; *fatr* is the fixed assets turnover ratio measured by net sales divided by net fixed assets; *mfer* is the manufacturing expenses ratio measured by power, fuel and water charges divided by net sales; *ccl* is the cash to current liabilities ratio measured by cash divided by current liabilities; *ctr* is creditors turnover ratio measured by net annual credit purchases divided by average trade creditors; *roce* is the return on capital employed measured by profit before income and taxes divided by total capital; *cf_rev* is cash flow to revenues measured by cash and cash equivalents at the end of the year divided by net sales; *qir* is the quality of income ratio measured by net cash flow from operating activities divided by profit before depreciation, interest, taxes and amortization; *tang* is the asset tangibility measured by total tangible assets divided by total assets; and *sz* is the size of the firm measured by natural logarithm of total assets.

Table 3: Time period descriptives

Year	Var	stdr	ltdr	tdr	ipsh	npsh	ipp	npp	mcev	mcr	pti	invtr	fa	indt	fatr	mfer	ccl	ctr	roce	cfrv	qir	tan	sz
2001	μ	0.20	0.30	0.49	19.12	17.88	71.71	28.43	0.68	0.30	13.13	3.15	29.92	12.96	4.64	0.07	0.45	6.06	1.97	0.20	0.82	0.34	10.45
2001	σ	0.16	0.22	0.26	1.81	1.81	19.87	20.50	0.48	0.24	10.66	3.58	15.64	20.06	6.86	0.09	0.50	5.04	1.78	0.23	0.90	0.21	1.66
2002	μ	0.20	0.29	0.48	19.22	17.92	72.71	27.46	0.78	0.27	13.60	4.52	30.87	12.97	4.64	0.07	0.39	6.45	2.06	0.19	0.78	0.32	10.58
2002	σ	0.16	0.21	0.23	1.87	1.85	19.17	19.97	0.51	0.24	12.12	8.49	15.43	19.62	8.03	0.09	0.37	6.61	2.02	0.20	0.61	0.21	1.63
2003	μ	0.23	0.25	0.48	19.19	17.96	71.30	28.91	0.84	0.32	14.14	4.63	31.15	18.88	6.25	0.06	0.38	6.46	2.57	0.20	0.74	0.32	10.62
2003	σ	0.19	0.23	0.23	1.87	1.86	20.38	21.09	0.47	0.27	11.77	6.78	15.87	23.34	11.25	0.09	0.35	5.65	3.11	0.25	0.83	0.22	1.66
2004	μ	0.23	0.22	0.44	19.21	17.96	70.95	27.89	0.92	0.30	15.95	3.71	32.15	23.67	6.72	0.06	0.51	5.91	2.99	0.31	0.97	0.30	10.73
2004	σ	0.20	0.21	0.22	1.88	1.83	20.20	18.82	0.38	0.25	14.43	4.57	15.87	22.60	12.35	0.08	0.50	5.28	2.90	0.59	0.90	0.27	1.62
2005	μ	0.23	0.21	0.42	19.31	18.05	70.85	28.09	0.99	0.31	17.15	3.50	32.12	27.65	6.52	0.05	0.61	6.35	3.91	0.31	0.62	0.33	10.87
2005	σ	0.19	0.20	0.23	1.90	1.82	20.28	19.02	0.42	0.29	14.69	4.44	16.18	28.76	11.13	0.08	0.67	6.87	3.69	0.31	0.87	0.21	1.56
2006	μ	0.22	0.20	0.41	19.29	18.06	70.70	27.97	0.98	0.33	17.19	3.56	33.12	29.61	7.88	0.07	0.75	6.74	4.29	0.45	0.58	0.31	10.98
2006	σ	0.18	0.19	0.23	1.94	1.83	20.97	18.89	0.36	0.29	14.61	4.63	16.18	30.42	15.71	0.12	0.92	6.37	4.77	0.93	0.45	0.22	1.54
2007	μ	0.21	0.19	0.39	19.30	18.07	70.49	27.87	0.99	0.32	17.88	3.56	34.12	32.16	6.36	0.06	0.77	7.46	5.23	0.53	0.56	0.29	11.13
2007	σ	0.16	0.18	0.21	1.95	1.83	21.14	18.88	0.39	0.28	16.22	5.08	16.18	32.26	7.70	0.09	0.90	7.64	5.44	1.15	0.52	0.19	1.54
2008	μ	0.19	0.20	0.37	19.40	18.19	70.46	28.08	1.09	0.33	18.82	3.10	34.44	36.06	7.78	0.06	0.98	6.28	5.77	0.64	0.54	0.26	11.27
2008	σ	0.15	0.20	0.23	1.88	1.78	20.74	18.62	0.56	0.27	15.56	4.40	16.59	38.74	12.62	0.08	1.07	5.75	5.65	1.48	0.88	0.22	1.60
2009	μ	0.19	0.21	0.38	19.52	18.32	70.51	28.04	1.04	0.33	17.89	3.49	35.44	31.92	5.93	0.06	0.82	7.07	5.69	0.47	0.49	0.27	11.47
2009	σ	0.14	0.21	0.23	1.96	1.68	20.72	18.61	0.56	0.27	16.61	5.48	16.59	37.29	6.15	0.08	0.90	6.95	6.91	0.53	0.90	0.17	1.52
2010	μ	0.19	0.22	0.39	19.54	18.43	69.65	29.01	0.95	0.33	17.99	3.02	36.44	27.05	5.48	0.05	0.83	6.29	5.09	0.51	0.50	0.22	11.61
2010	σ	0.15	0.21	0.23	1.92	1.72	20.74	18.78	0.35	0.27	16.93	5.47	16.59	27.56	6.40	0.07	0.93	5.56	4.74	0.55	0.67	0.27	1.53
2011	μ	0.26	0.21	0.45	19.71	18.66	69.13	29.49	0.97	0.34	17.70	3.56	37.44	29.00	5.79	0.05	0.83	6.88	5.46	0.40	0.41	0.27	11.77
2011	σ	0.20	0.21	0.27	1.94	1.55	20.45	18.44	0.45	0.28	16.49	7.60	16.59	29.12	7.57	0.07	0.92	5.66	5.65	0.48	0.68	0.18	1.52
2012	μ	0.26	0.23	0.46	19.74	18.73	68.61	30.04	0.95	0.35	17.54	3.81	38.44	27.97	6.75	0.06	0.74	7.06	5.53	0.36	0.30	0.25	11.91
2012	σ	0.20	0.21	0.28	1.95	1.59	20.21	18.24	0.45	0.29	17.78	7.41	16.59	31.63	11.56	0.09	0.90	6.02	6.45	0.50	0.59	0.19	1.50
2013	μ	0.26	0.25	0.49	19.82	18.95	67.75	31.11	0.94	0.32	16.62	3.70	39.44	26.37	6.57	0.06	0.71	6.78	5.37	0.36	0.51	0.25	11.97
2013	σ	0.19	0.22	0.30	1.89	1.50	19.91	18.24	0.56	0.27	16.92	5.22	16.59	28.18	8.91	0.09	0.81	5.40	5.65	0.55	1.08	0.19	1.51
2014	μ	0.25	0.25	0.47	19.81	19.09	67.07	31.80	0.97	0.34	15.46	3.57	40.44	23.26	5.22	0.06	0.66	7.23	5.27	0.51	0.59	0.28	12.05
2014	σ	0.17	0.22	0.29	1.87	1.40	19.50	17.82	0.71	0.29	16.79	5.69	16.59	25.89	5.57	0.08	0.72	5.87	5.46	1.19	0.90	0.18	1.53
2015	μ	0.23	0.24	0.45	19.80	19.12	67.00	31.94	0.92	0.34	14.69	3.27	41.44	25.15	4.83	0.06	0.69	6.86	4.93	0.74	0.59	0.28	12.09
2015	σ	0.18	0.23	0.29	1.86	1.42	19.55	17.94	0.54	0.28	16.42	4.86	16.59	26.62	5.78	0.07	0.77	5.53	5.93	1.90	0.84	0.21	1.51
2016	μ	0.24	0.25	0.46	19.81	19.17	66.59	32.64	0.95	0.32	13.87	2.70	42.44	23.03	5.44	0.06	0.65	6.61	4.71	0.64	0.46	0.28	12.14
2016	σ	0.16	0.23	0.30	1.84	1.40	19.65	18.18	0.56	0.27	16.04	2.84	16.59	25.51	14.22	0.07	0.61	5.57	5.38	1.62	0.74	0.21	1.50
All years	μ	0.22	0.23	0.44	19.49	18.41	69.72	29.30	0.94	0.32	16.23	3.55	35.59	25.48	6.05	0.06	0.67	6.65	4.43	0.43	0.59	0.29	11.35
All years	σ	0.17	0.21	0.25	1.90	1.68	20.22	18.88	0.48	0.27	15.25	5.41	16.29	27.97	9.49	0.08	0.74	5.99	4.72	0.78	0.77	0.21	1.56

Table 4: Cross-section descriptives of the dependent variable

	STDR		LTDR		TDR		In_IP_SH		In_NP_SH		IP_P		NP_P	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
BALMLAWRIE	0.35	0.06	0.11	0.13	0.44	0.13	16.12	0	16.59	0.43	61.80	0	38.20	0
BEL	0.46	0.07	0.01	0.02	0.47	0.08	17.99	0.27	16.85	0.28	75.70	0	24.29	0
BEML	0.41	0.13	0.12	0.08	0.53	0.10	16.93	0	16.64	0.15	57.18	0	42.82	0
BHEL	0.36	0.08	0.02	0.03	0.38	0.09	19.81	1.00	19.10	1.06	66.85	0	33.15	0
BPCL	0.41	0.13	0.29	0.08	0.68	0.12	19.31	0.30	18.86	0.48	60.91	0	39.09	0
CHENNPETRO	0.38	0.21	0.34	0.08	0.65	0.15	18.16	0	17.88	0.38	51.89	0	42.45	0
COALINDIA	0.19	0.05	0.11	0.08	0.29	0.12	22.44	0.04	20.36	0.25	90.00	0	10.00	0
CONCOR	0.08	0.01	0.03	0.02	0.10	0.03	17.94	0.45	17.43	0.48	62.85	0.52	37.15	0.52
ENGINEERSIN	0.26	0.09	0.02	0.02	0.26	0.09	18.33	0.79	16.54	1.40	83.95	9.94	16.04	9.95
GAIL	0.15	0.01	0.12	0.06	0.27	0.06	20.23	0.18	19.83	0.28	59.02	4.15	39.90	3.91
GMDCLTD	0.09	0.03	0.19	0.18	0.27	0.20	18.23	1.15	17.18	1.15	74.00	0	26.00	0
GNFC	0.16	0.08	0.25	0.12	0.41	0.18	17.97	0	18.29	0.05	42.13	1.26	57.78	1.19
GSFC	0.19	0.05	0.24	0.13	0.44	0.15	17.62	0.72	18.12	0.72	37.84	0	62.16	0.01
HINDCOPPER	0.20	0.12	0.29	0.32	0.42	0.35	20.33	0.38	15.92	1.38	97.25	3.86	2.74	3.86
HINDPETRO	0.36	0.11	0.31	0.13	0.66	0.19	18.97	0	18.93	0	51.07	0.05	48.93	0.05
IFCI	0.09	0.05	0.72	0.12	0.80	0.14	19.98	0.40	20.36	0.08	55.53	0	44.47	0
IGL	0.14	0.05	0.10	0.06	0.23	0.12	17.96	0	18.16	0	45.00	0	55.00	0
IOC	0.31	0.09	0.29	0.06	0.60	0.12	20.87	0.42	19.58	0.67	78.52	5.10	22.11	6.75
MGL	0.15	0.03	0.09	0.12	0.25	0.11	19.96	0.12	19.22	0.07	52.50	6.11	48.06	5.09
MMTC	0.59	0.18	0.20	0.13	0.72	0.13	18.83	1.47	14.35	2.34	90.00	0	10.00	0
MOIL	0.12	0.06	0.06	0.02	0.12	0.06	18.72	0	17.33	0	80.00	0	20.00	0
MRPL	0.36	0.17	0.33	0.21	0.70	0.14	21.03	0.33	19.18	0.26	88.57	0.01	11.43	0.01
MTNL	0.26	0.09	0.25	0.20	0.43	0.27	19.69	0	19.37	0.07	56.25	0	41.04	2.82
NATIONALUM	0.12	0.04	0.12	0.07	0.17	0.07	20.65	0.67	18.85	0.83	85.63	2.72	14.39	2.75
NBCC	0.72	0.07	0.12	0.09	0.78	0.14	18.50	0	16.30	0	90.00	0	10.00	0
NHPC	0.06	0.02	0.34	0.04	0.40	0.03	23.07	0.04	21.23	0.03	86.29	0.16	13.72	0.16
NLCINDIA	0.08	0.03	0.16	0.04	0.24	0.06	21.17	0.02	18.58	0.18	93.56	0	6.44	0
NMDC	0.05	0.02	0.07	0.06	0.05	0.02	20.32	1.69	17.25	2.82	98.38	0	1.62	0.01
NTPC	0.08	0.02	0.30	0.06	0.38	0.07	22.66	0.08	20.88	0.40	87.31	2.56	12.69	2.56
OIL	0.10	0.05	0.08	0.09	0.18	0.11	19.25	0.34	18.09	0.59	75.78	4.74	24.22	4.74
ONGC	0.11	0.06	0.05	0.04	0.13	0.05	21.55	0.78	20.46	1.01	74.42	5.33	25.58	5.33
PETRONET	0.15	0.07	0.36	0.09	0.51	0.09	19.74	0	19.74	0	50.00	0	50.00	0
PFC	0.07	0.06	0.76	0.05	0.83	0.11	20.73	0.04	18.94	0.55	84.76	7.69	15.24	7.69
PFS	0.14	0.09	0.45	0.24	0.54	0.38	19.64	0	19.23	0	60.00	0	40.00	0
POWERGRID	0.10	0.03	0.55	0.02	0.65	0.04	21.96	0.08	20.59	0.58	86.36	0	13.64	0
PTC	0.27	0.11	0.06	0.05	0.27	0.11	17.69	0	18.96	0.42	32.00	0	68.00	0
RCF	0.25	0.15	0.25	0.08	0.50	0.20	20.01	0.06	17.78	0.44	92.50	0	7.50	0.01
RECLTD	0.11	0.06	0.80	0.01	0.91	0.06	20.31	0.05	19.47	0.35	69.20	7.42	30.80	7.42
SAIL	0.21	0.06	0.29	0.16	0.49	0.18	21.96	0.05	20.30	0.21	84.37	2.60	15.63	2.60
SCI	0.15	0.03	0.31	0.12	0.45	0.13	19.39	0.16	18.31	0.53	73.98	8.18	26.02	8.19
SJVN	0.07	0.04	0.22	0.10	0.29	0.10	22.04	0	19.84	0	89.97	0	10.03	0
SMLISUZU	0.44	0.13	0.16	0.11	0.60	0.14	14.42	0.40	15.57	0.24	18.73	7.18	47.92	4.87
TNPL	0.20	0.11	0.37	0.09	0.58	0.18	17.01	0	17.62	0	35.25	0.11	64.75	0.10
All years	0.22	0.08	0.24	0.09	0.44	0.13	19.52	0.29	18.47	0.49	69.47	1.85	29.56	1.87

Full model of Leverage

VARIABLES	Short-term debt ratio (<i>stdr</i>)				Long-term debt ratio (<i>ltdr</i>)				Total-debt ratio (<i>tdr</i>)			
	OLS1	FE1	TFE1	RE1	OLS2	FE2	TFE2	RE2	OLS3	FE3	TFE3	RE3
in_ip_sh	-0.0206** (0.00829)	0.0190 (0.0122)	0.0154 (0.0121)	0.000444 (0.0105)	-0.0412*** (0.00919)	-0.0509*** (0.0123)	-0.0436*** (0.0124)	-0.0409*** (0.0112)	-0.0692*** (0.0118)	-0.0457*** (0.0172)	-0.0416** (0.0172)	-0.0497*** (0.0149)
ip_p	-0.000618 (0.000792)	0.00179 (0.00131)	0.00197 (0.00129)	-0.00154 (0.000989)	0.00500*** (0.000878)	-0.000901 (0.00132)	-0.00142 (0.00132)	0.00293*** (0.00106)	0.00522*** (0.00112)	0.00159 (0.00185)	0.00136 (0.00183)	0.00305** (0.00140)
in_np_sh	-0.00180 (0.00684)	-0.0183** (0.00890)	-0.0184** (0.00889)	-0.0105 (0.00818)	0.0344*** (0.00759)	0.0179** (0.00891)	0.0100 (0.00910)	0.0139 (0.00862)	0.0397*** (0.00971)	0.00912 (0.0125)	0.00160 (0.0126)	0.0109 (0.0115)
np_p	-0.00113 (0.000725)	-0.00120 (0.000882)	-0.00143* (0.000866)	-0.00145* (0.000838)	0.00244*** (0.000804)	0.000510 (0.000883)	5.63e-05 (0.000886)	0.00105 (0.000877)	0.00154 (0.00103)	-6.78e-05 (0.00124)	-0.000681 (0.00123)	0.000366 (0.00118)
mc_ev	0.0458*** (0.0114)	0.00170 (0.0110)	0.00435 (0.0109)	0.0172 (0.0109)	-0.187*** (0.0126)	-0.142*** (0.0110)	-0.138*** (0.0111)	-0.153*** (0.0112)	-0.160*** (0.0161)	-0.153*** (0.0154)	-0.148*** (0.0154)	-0.155*** (0.0152)
mcr	0.0572*** (0.0192)	0.118*** (0.0378)	0.115*** (0.0371)	0.0912*** (0.0293)	0.0892*** (0.0213)	0.106*** (0.0378)	0.121*** (0.0379)	0.0964*** (0.0323)	0.136*** (0.0273)	0.191*** (0.0531)	0.202*** (0.0526)	0.174*** (0.0419)
pti	-0.00161*** (0.000400)	-0.00188*** (0.000603)	-0.00190*** (0.000596)	-0.00233*** (0.000505)	0.000939** (0.000444)	-0.00225*** (0.000604)	-0.00183*** (0.000610)	-0.000467 (0.000544)	-0.00113** (0.000567)	-0.00478*** (0.000848)	-0.00445*** (0.000845)	-0.00316*** (0.000716)
invtttr	0.0123*** (0.000959)	0.00887*** (0.000933)	0.00863*** (0.000913)	0.0100*** (0.000921)	0.00832*** (0.00106)	0.00632*** (0.000934)	0.00620*** (0.000934)	0.00639*** (0.000950)	0.0148*** (0.00136)	0.00846*** (0.00131)	0.00805*** (0.00129)	0.00996*** (0.00129)
fa	0.00172*** (0.000303)	0.00581*** (0.00148)	0.00527*** (0.00163)	0.00275*** (0.000539)	-0.00182*** (0.000336)	-0.00979*** (0.00148)	-0.00989*** (0.00167)	-0.00328*** (0.000638)	-0.000383 (0.000429)	-0.00678*** (0.00208)	-0.00714*** (0.00232)	-0.00112 (0.000785)
indebt	0.00101*** (0.000221)	0.000535** (0.000214)	0.000590*** (0.000211)	0.000774*** (0.000209)	-0.00288*** (0.000245)	-0.00131*** (0.000215)	-0.00119*** (0.000216)	-0.00182*** (0.000216)	-0.00230*** (0.000313)	-0.000874*** (0.000301)	-0.000724** (0.000300)	-0.00136*** (0.000293)
fatr	0.00435*** (0.000513)	0.00171*** (0.000512)	0.00170*** (0.000504)	0.00258*** (0.000501)	-0.00160*** (0.000569)	-0.00164*** (0.000513)	-0.00150*** (0.000516)	-0.00184*** (0.000518)	0.00370*** (0.000727)	0.00108 (0.000719)	0.00114 (0.000714)	0.00160** (0.000702)
mfer	-0.0702 (0.0522)	-0.0195 (0.0751)	-0.00209 (0.0736)	-0.0586 (0.0673)	0.410*** (0.0579)	0.396*** (0.0753)	0.412*** (0.0752)	0.432*** (0.0717)	0.238*** (0.0741)	0.364*** (0.106)	0.396*** (0.104)	0.332*** (0.0952)
ccl	-0.0561*** (0.00649)	-0.0292*** (0.00687)	-0.0260*** (0.00677)	-0.0332*** (0.00669)	0.0202*** (0.00719)	0.0212*** (0.00688)	0.0235*** (0.00693)	0.0177** (0.00694)	-0.0415*** (0.00920)	-0.00791 (0.00965)	-0.00193 (0.00960)	-0.0181* (0.00938)
ctr	-0.00166** (0.000775)	-0.00489*** (0.000894)	-0.00501*** (0.000875)	-0.00372*** (0.000846)	-0.00608*** (0.000859)	-0.00443*** (0.000895)	-0.00463*** (0.000895)	-0.00541*** (0.000883)	-0.00648*** (0.00110)	-0.00912*** (0.00126)	-0.00942*** (0.00124)	-0.00877*** (0.00119)
roce	-0.00627*** (0.00121)	-0.00223* (0.00114)	-0.00180 (0.00111)	-0.00337*** (0.00113)	0.00622*** (0.00134)	0.00156 (0.00114)	0.00178 (0.00114)	0.00284** (0.00116)	0.00236 (0.00171)	0.00214 (0.00160)	0.00273* (0.00158)	0.00265* (0.00158)
cf_rev	0.00933** (0.00461)	0.00239 (0.00406)	0.00557 (0.00404)	0.00386 (0.00404)	0.0167*** (0.00511)	0.00134 (0.00407)	0.00150 (0.00413)	0.00562 (0.00415)	0.0279*** (0.00653)	0.00815 (0.00571)	0.0110* (0.00572)	0.0135** (0.00564)
qir	0.0132** (0.00532)	0.00776* (0.00437)	0.00855** (0.00430)	0.00742* (0.00449)	-0.0302*** (0.00590)	-0.0176*** (0.00438)	-0.0196*** (0.00440)	-0.0186*** (0.00458)	-0.0156** (0.00754)	-0.00781 (0.00614)	-0.00893 (0.00610)	-0.00924 (0.00626)
tang	-0.0326	-0.0465	-0.0553*	-0.0555**	-0.0281	0.0851***	0.0795**	0.0562*	-0.0571	0.00786	-0.0106	-0.0244

	(0.0248)	(0.0304)	(0.0301)	(0.0280)	(0.0275)	(0.0305)	(0.0308)	(0.0295)	(0.0352)	(0.0428)	(0.0427)	(0.0394)
sz	0.0180*** (0.00469)	0.00229 (0.0104)	-0.00378 (0.0104)	0.0102* (0.00600)	0.0132** (0.00521)	0.0766*** (0.0105)	0.0799*** (0.0106)	0.0421*** (0.00657)	0.0279*** (0.00666)	0.0798*** (0.0147)	0.0783*** (0.0147)	0.0454*** (0.00855)
2.period			-0.0146 (0.0163)				0.00619 (0.0167)				-0.00515 (0.0232)	
3.period			0.00319 (0.0158)				-0.00999 (0.0162)				0.00455 (0.0224)	
4.period			0.00498 (0.0155)				-0.0140 (0.0159)				-0.00113 (0.0220)	
5.period			0.00888 (0.0151)				-0.0247 (0.0155)				-0.00369 (0.0214)	
6.period			-0.00721 (0.0149)				-0.0405*** (0.0153)				-0.0325 (0.0212)	
7.period			-0.0103 (0.0148)				-0.0356** (0.0151)				-0.0333 (0.0210)	
8.period			-0.0327** (0.0148)				-0.0277* (0.0151)				-0.0544*** (0.0210)	
9.period			-0.0348** (0.0144)				-0.0292** (0.0148)				-0.0584*** (0.0205)	
10.period			-0.0285** (0.0143)				-0.0378** (0.0147)				-0.0657*** (0.0203)	
11.period			0.0331** (0.0144)				-0.0435*** (0.0148)				-0.00480 (0.0205)	
12.period			0.0303** (0.0147)				-0.0342** (0.0151)				-0.00738 (0.0209)	
13.period			0.0305** (0.0147)				-0.00826 (0.0150)				0.0286 (0.0208)	
14.period			0.0200 (0.0149)				-0.00760 (0.0152)				0.0169 (0.0211)	
15.period			-0.00401 (0.0152)				-0.0151 (0.0156)				-0.0167 (0.0216)	
16.period			-				-				-	
Constant	0.419*** (0.0811)	-0.124 (0.178)	0.0237 (0.182)	0.343*** (0.108)	0.0750 (0.0899)	0.547*** (0.179)	0.569*** (0.187)	0.313*** (0.119)	0.513*** (0.115)	0.567** (0.251)	0.686*** (0.259)	0.686*** (0.154)
Observations	663	663	663	663	663	663	663	663	663	663	663	663
R-squared	0.708	0.325	0.377		0.752	0.567	0.583		0.716	0.478	0.509	
N		43	43	43		43	43	43		43	43	43
Firm FE		YES	YES			YES	YES			YES	YES	
Year FE			YES				YES				YES	

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The dependent variable elicits leverage measures such as: *stdr* is the short-term debt ratio measured as short-term debt divided by total assets; *ltdr* is the long-term debt ratio measured as long-term debt divided by total assets; *tdr* is the total-debt ratio measured as the sum of short-term debt and long-term debt divided by total assets. All other regressors remain the same. Both the pooled ordinary least squares model and the fixed effects model are corrected for heteroscedasticity effects using both Breusch-Pagan and White methods, correspondingly. The figures as shown in the parentheses represent standard errors.

Full model of Ownership Structure

VAR.	<i>ln_ip_sh</i>				<i>ln_np_sh</i>				<i>ip_p</i>				<i>np_p</i>			
	OLS1	FE1	TFE1	RE1	OLS2	FE2	TFE2	RE2	OLS3	FE3	TFE3	RE3	OLS4	FE4	TFE4	RE4
stdr	-2.071*** (0.783)	0.286 (0.417)	0.152 (0.419)	0.288 (0.438)	-1.535** (0.780)	-0.965 (0.607)	-1.082* (0.600)	-0.689 (0.644)	-22.94* (12.13)	7.336** (2.898)	7.831*** (2.933)	6.524** (3.010)	12.50 (11.63)	-13.51*** (4.281)	-14.24*** (4.288)	-11.83*** (4.431)
ltdr	-0.417 (0.773)	-0.573 (0.417)	-0.683 (0.419)	-0.572 (0.440)	0.0982 (0.770)	-0.652 (0.608)	-0.900 (0.600)	-0.510 (0.646)	-7.324 (11.98)	-2.739 (2.901)	-2.820 (2.932)	-2.544 (3.016)	9.227 (11.48)	0.267 (4.286)	-0.683 (4.286)	-0.0938 (4.443)
tdr	0.279 (0.716)	-0.509 (0.382)	-0.507 (0.383)	-0.626 (0.402)	0.421 (0.714)	0.0151 (0.557)	-0.0717 (0.548)	-0.346 (0.591)	14.29 (11.10)	-0.892 (2.658)	-0.275 (2.682)	-0.308 (2.761)	-8.506 (10.64)	4.394 (3.927)	3.528 (3.921)	3.213 (4.064)
mc_ev	0.487*** (0.129)	-0.0527 (0.0746)	-0.0300 (0.0754)	-0.0746 (0.0783)	0.337*** (0.128)	-0.0156 (0.109)	0.0121 (0.108)	-0.0311 (0.114)	7.052*** (1.995)	0.0750 (0.518)	0.0680 (0.528)	0.199 (0.538)	-5.358*** (1.913)	-0.0179 (0.766)	0.0782 (0.772)	-0.274 (0.792)
mcr	-0.665*** (0.183)	-0.525** (0.225)	-0.446* (0.227)	-0.691*** (0.221)	0.869*** (0.182)	-0.738** (0.328)	-0.550* (0.325)	0.112 (0.263)	-19.22*** (2.838)	-3.926** (1.565)	-4.214*** (1.591)	-5.010*** (1.607)	12.53*** (2.721)	3.167 (2.313)	4.159* (2.326)	5.313** (2.318)
pti	0.0368*** (0.00366)	0.00214 (0.00366)	0.00318 (0.00368)	0.00638* (0.00364)	0.0237*** (0.00365)	0.00850 (0.00533)	0.0113** (0.00527)	0.0123*** (0.00457)	0.295*** (0.0568)	0.0226 (0.0254)	0.0249 (0.0258)	0.0404 (0.0261)	-0.323*** (0.0544)	-0.0414 (0.0376)	-0.0308 (0.0377)	-0.0811** (0.0379)
invtttr	0.0531*** (0.0111)	0.0144** (0.00639)	0.0162** (0.00642)	0.0132** (0.00673)	-0.0403*** (0.0111)	0.00305 (0.00931)	0.00669 (0.00918)	-0.00982 (0.00984)	0.755*** (0.172)	-0.00169 (0.0445)	-0.0119 (0.0449)	0.00484 (0.0462)	-0.796*** (0.165)	-0.0383 (0.0657)	-0.0194 (0.0656)	-0.0537 (0.0680)
fa	-0.00729** (0.00283)	0.0242*** (0.00889)	0.0149 (0.0101)	-0.0133** (0.00600)	-0.0145*** (0.00282)	0.0636*** (0.0129)	0.0438*** (0.0145)	-0.0165*** (0.00499)	0.263*** (0.0439)	-0.436*** (0.0618)	-0.395*** (0.0707)	-0.359*** (0.0574)	-0.211*** (0.0421)	0.389*** (0.0913)	0.320*** (0.103)	0.239*** (0.0733)
indebt	0.00600** (0.00239)	0.00123 (0.00132)	0.00181 (0.00132)	0.00154 (0.00139)	0.00899*** (0.00238)	0.00145 (0.00192)	0.00262 (0.00190)	0.00285 (0.00205)	0.0422 (0.0370)	0.0140 (0.00917)	0.0121 (0.00927)	0.0127 (0.00953)	0.0151 (0.0355)	-0.00522 (0.0135)	0.000918 (0.0135)	-0.00243 (0.0140)
fatr	0.00720 (0.00538)	0.00924*** (0.00309)	0.0103*** (0.00311)	0.00868*** (0.00326)	0.00621 (0.00536)	0.0127*** (0.00451)	0.0147*** (0.00445)	0.00795* (0.00475)	0.129 (0.0834)	0.0113 (0.0215)	0.0103 (0.0218)	0.0148 (0.0224)	-0.130 (0.0799)	-0.0129 (0.0318)	-0.00561 (0.0318)	-0.0207 (0.0329)
mfer	1.048* (0.537)	0.675 (0.456)	0.825* (0.458)	0.813* (0.470)	0.0480 (0.535)	0.482 (0.665)	0.862 (0.656)	0.984 (0.636)	9.738 (8.324)	0.909 (3.173)	0.460 (3.209)	0.282 (3.286)	-4.197 (7.980)	-1.883 (4.688)	0.0387 (4.691)	-0.497 (4.809)
ccl	-0.139** (0.0667)	-0.0910** (0.0414)	-0.0827** (0.0419)	-0.0771* (0.0433)	-0.309*** (0.0664)	-0.204*** (0.0602)	-0.170*** (0.0599)	-0.167*** (0.0625)	2.343** (1.033)	0.283 (0.288)	0.212 (0.293)	0.240 (0.299)	-2.808*** (0.991)	-0.366 (0.425)	-0.138 (0.428)	-0.277 (0.439)
ctr	-0.0152** (0.00757)	0.00231 (0.00554)	-2.53e-05 (0.00557)	6.48e-05 (0.00577)	0.0364*** (0.00754)	-0.00736 (0.00806)	-0.0126 (0.00797)	0.000170 (0.00810)	-0.750*** (0.117)	0.0724* (0.0385)	0.0825** (0.0390)	0.0623 (0.0400)	0.931*** (0.112)	-0.0606 (0.0569)	-0.0883 (0.0570)	-0.0322 (0.0587)
roce	-0.135*** (0.0112)	-0.0157** (0.00685)	-0.0135** (0.00686)	-0.0286*** (0.00712)	-0.0742*** (0.0111)	-0.0195* (0.00998)	-0.0145 (0.00982)	-0.0378*** (0.0102)	-1.258*** (0.173)	-0.0264 (0.0476)	-0.0346 (0.0480)	-0.0322 (0.0493)	1.029*** (0.166)	0.0167 (0.0704)	0.0451 (0.0702)	0.0342 (0.0723)
cf_rev	-0.0722 (0.0459)	-0.0316 (0.0240)	-0.0143 (0.0245)	-0.0416* (0.0252)	0.0389 (0.0457)	-0.103*** (0.0350)	-0.0823** (0.0350)	-0.0835** (0.0370)	-1.846*** (0.712)	0.215 (0.167)	0.183 (0.171)	0.226 (0.173)	2.044*** (0.682)	-0.211 (0.247)	-0.0943 (0.250)	-0.225 (0.255)
qir	0.0131 (0.0537)	-0.0246 (0.0263)	-0.0303 (0.0266)	-0.0305 (0.0279)	0.0132 (0.0535)	0.00946 (0.0383)	-0.0133 (0.0381)	0.000653 (0.0418)	-0.0297 (0.833)	-0.357* (0.183)	-0.333* (0.186)	-0.344* (0.190)	0.445 (0.798)	0.324 (0.270)	0.235 (0.273)	0.303 (0.281)
tang	0.661*** (0.237)	0.299 (0.182)	0.302 (0.185)	0.438** (0.187)	1.688*** (0.236)	0.410 (0.265)	0.338 (0.264)	0.847*** (0.256)	-8.944** (3.672)	-1.901 (1.265)	-1.529 (1.292)	-2.029 (1.310)	8.893** (3.520)	0.329 (1.869)	-0.216 (1.888)	0.675 (1.916)
sz	0.896***	0.235***	0.210***	0.540***	0.687***	0.324***	0.335***	0.790***	5.132***	0.383	0.331	0.0305	-4.239***	-0.570	-0.446	0.0785

	(0.0270)	(0.0641)	(0.0656)	(0.0456)	(0.0269)	(0.0933)	(0.0939)	(0.0424)	(0.418)	(0.445)	(0.459)	(0.418)	(0.401)	(0.658)	(0.671)	(0.542)
2.period			-0.0553 (0.0996)				-0.112 (0.143)				0.678 (0.697)				-0.552 (1.019)	
3.period			-0.0902 (0.0964)				-0.168 (0.138)				0.400 (0.675)				-0.195 (0.986)	
4.period			-0.138 (0.0941)				-0.288** (0.135)				0.154 (0.659)				-1.363 (0.963)	
5.period			-0.163* (0.0917)				-0.356*** (0.131)				0.161 (0.642)				-1.367 (0.938)	
6.period			-0.232** (0.0905)				-0.440*** (0.130)				0.322 (0.634)				-1.838** (0.926)	
7.period			-0.234*** (0.0892)				-0.476*** (0.128)				0.348 (0.624)				-2.179** (0.913)	
8.period			-0.199** (0.0892)				-0.486*** (0.128)				0.896 (0.625)				-2.719*** (0.913)	
9.period			-0.126 (0.0870)				-0.446*** (0.125)				1.403** (0.609)				-2.971*** (0.890)	
10.period			-0.113 (0.0870)				-0.389*** (0.125)				0.891 (0.609)				-2.260** (0.890)	
11.period			0.00990 (0.0883)				-0.195 (0.126)				0.279 (0.618)				-1.323 (0.904)	
12.period			-0.00539 (0.0897)				-0.222* (0.128)				0.123 (0.627)				-1.006 (0.917)	
13.period			0.0639 (0.0895)				-0.0640 (0.128)				-0.284 (0.626)				-0.293 (0.916)	
14.period			0.0530 (0.0907)				0.0563 (0.130)				-0.325 (0.635)				-0.105 (0.928)	
15.period			0.0166 (0.0928)				0.0224 (0.133)				0.0819 (0.649)				-0.449 (0.949)	
16.period			-				-				-				-	
Constant	9.354*** (0.407)	16.32*** (0.478)	16.95*** (0.541)	14.29*** (0.429)	9.949*** (0.406)	12.99*** (0.696)	13.72*** (0.774)	10.20*** (0.489)	4.544 (6.311)	80.70*** (3.324)	79.24*** (3.787)	81.62*** (3.974)	81.03*** (6.050)	23.48*** (4.910)	25.59*** (5.537)	21.75*** (4.913)
Obs.	663	663	663	663	663	663	663	663	663	663	663	663	663	663	663	663
R-squared	0.753	0.276	0.299		0.705	0.343	0.385		0.472	0.269	0.283		0.442	0.120	0.154	
N		43	43	43		43	43	43		43	43	43		43	43	43
Firm FE		YES	YES			YES	YES			YES	YES			YES	YES	
Year FE			YES				YES				YES				YES	

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The dependent variable elicits ownership structure measures such as: *in_ip_sh* is the natural logarithm of the number of shares owned by Indian promoters (i.e. total stakes of individuals & hindu undivided family; central & state government promoters; corporate bodies; financial institutions & banks; and other indian promoters); *in_np_sh* is the natural logarithm of the number of shares owned by non promoters (i.e. non-promoter institutions & non-promoter non-institutions); *ip_p* is the percentage of ownership stakes held by indian promoters; *np_p* is the percentage of ownership stakes held by non-promoters. Independent variables also include short-term debt ratio; long-term debt ratio; and total debt ratio. The remaining regressors as included earlier remain the same. Both the pooled ordinary least squares model and the fixed effects model are corrected for heteroscedasticity effects using both Breusch-Pagan and White methods, correspondingly. The figures as shown in the parentheses represent standard errors.

Reduced-form model of Leverage

VAR.	Short-term debt ratio (<i>stdr</i>)				Long-term debt ratio (<i>ltdr</i>)				Total debt ratio (<i>tldr</i>)			
	OLS1	FE1	TFE1	RE1	OLS2	FE2	TFE2	RE2	OLS3	FE3	TFE3	RE3
mc_ev	0.0391*** (0.0114)	0.00463 (0.0111)	0.00566 (0.0111)	0.0150 (0.0109)	-0.185*** (0.0128)	-0.148*** (0.0112)	-0.144*** (0.0113)	-0.154*** (0.0113)	-0.164*** (0.0163)	-0.156*** (0.0155)	-0.153*** (0.0156)	-0.158*** (0.0154)
mcr	0.0697*** (0.0183)	0.113*** (0.0378)	0.108*** (0.0373)	0.102*** (0.0290)	0.0856*** (0.0205)	0.128*** (0.0380)	0.147*** (0.0381)	0.105*** (0.0323)	0.143*** (0.0262)	0.207*** (0.0527)	0.220*** (0.0526)	0.180*** (0.0411)
pti	-0.00232*** (0.000363)	-0.00191*** (0.000608)	-0.00203*** (0.000601)	-0.00245*** (0.000496)	0.000955** (0.000407)	-0.00235*** (0.000612)	-0.00198*** (0.000614)	-0.000876 (0.000542)	-0.00175*** (0.000520)	-0.00489*** (0.000849)	-0.00467*** (0.000847)	-0.00352*** (0.000702)
invtr	0.0122*** (0.000914)	0.00947*** (0.000928)	0.00921*** (0.000913)	0.0103*** (0.000902)	0.00668*** (0.00102)	0.00554*** (0.000934)	0.00562*** (0.000933)	0.00591*** (0.000938)	0.0129*** (0.00131)	0.00800*** (0.00130)	0.00778*** (0.00129)	0.00962*** (0.00128)
fa	0.00205*** (0.000277)	0.00426*** (0.00144)	0.00377** (0.00163)	0.00306*** (0.000556)	-0.00130*** (0.000311)	-0.00992*** (0.00145)	-0.0101*** (0.00166)	-0.00302*** (0.000678)	0.000581 (0.000397)	-0.00841*** (0.00201)	-0.00880*** (0.00229)	-0.000294 (0.000785)
indebt	0.000875*** (0.000222)	0.000617*** (0.000216)	0.000637*** (0.000214)	0.000717*** (0.000209)	-0.00276*** (0.000249)	-0.00145*** (0.000217)	-0.00133*** (0.000218)	-0.00183*** (0.000217)	-0.00229*** (0.000318)	-0.000944*** (0.000301)	-0.000818*** (0.000301)	-0.00144*** (0.000296)
fatr	0.00443*** (0.000517)	0.00178*** (0.000514)	0.00171*** (0.000507)	0.00242*** (0.000500)	-0.00138** (0.000579)	-0.00200*** (0.000517)	-0.00190*** (0.000518)	-0.00198*** (0.000520)	0.00407*** (0.000740)	0.000758 (0.000717)	0.000742 (0.000714)	0.00141** (0.000708)
mfer	-0.0978* (0.0527)	-0.0203 (0.0763)	-0.00889 (0.0751)	-0.0611 (0.0684)	0.427*** (0.0590)	0.387*** (0.0768)	0.403*** (0.0767)	0.411*** (0.0731)	0.231*** (0.0754)	0.353*** (0.107)	0.380*** (0.106)	0.310*** (0.0968)
ccl	-0.0531*** (0.00650)	-0.0277*** (0.00691)	-0.0249*** (0.00686)	-0.0311*** (0.00669)	0.0211*** (0.00727)	0.0233*** (0.00696)	0.0266*** (0.00701)	0.0194*** (0.00697)	-0.0373*** (0.00929)	-0.00461 (0.00965)	0.00197 (0.00966)	-0.0152 (0.00948)
ctr	-0.00207*** (0.000738)	-0.00457*** (0.000902)	-0.00465*** (0.000887)	-0.00396*** (0.000846)	-0.00598*** (0.000826)	-0.00494*** (0.000908)	-0.00507*** (0.000906)	-0.00560*** (0.000891)	-0.00688*** (0.00106)	-0.00942*** (0.00126)	-0.00962*** (0.00125)	-0.00925*** (0.00120)
roce	-0.00392*** (0.00111)	-0.00242** (0.00115)	-0.00202* (0.00113)	-0.00285*** (0.00109)	0.00591*** (0.00124)	0.00228** (0.00115)	0.00247** (0.00115)	0.00375*** (0.00113)	0.00414*** (0.00159)	0.00276* (0.00160)	0.00329** (0.00159)	0.00407*** (0.00154)
cf_rev	0.00991** (0.00465)	0.00440 (0.00409)	0.00765* (0.00408)	0.00471 (0.00401)	0.0180*** (0.00520)	0.000877 (0.00411)	0.00110 (0.00417)	0.00590 (0.00415)	0.0297*** (0.00665)	0.00919 (0.00570)	0.0121** (0.00574)	0.0149*** (0.00568)
qir	0.0131** (0.00540)	0.00664 (0.00443)	0.00787* (0.00439)	0.00726 (0.00447)	-0.0312*** (0.00605)	-0.0166*** (0.00446)	-0.0188*** (0.00448)	-0.0180*** (0.00460)	-0.0170** (0.00773)	-0.00768 (0.00618)	-0.00881 (0.00618)	-0.00913 (0.00634)
tang	-0.0560** (0.0242)	-0.0553* (0.0308)	-0.0628** (0.0306)	-0.0627** (0.0280)	-0.0213 (0.0271)	0.0831*** (0.0310)	0.0751** (0.0312)	0.0441 (0.0297)	-0.0722** (0.0347)	-0.00336 (0.0429)	-0.0240 (0.0431)	-0.0514 (0.0397)
sz	-0.000211 (0.00272)	0.000650 (0.0105)	-0.00656 (0.0105)	0.00216 (0.00475)	0.0161*** (0.00304)	0.0735*** (0.0106)	0.0773*** (0.0107)	0.0283*** (0.00558)	0.0144*** (0.00388)	0.0755*** (0.0147)	0.0739*** (0.0148)	0.0220*** (0.00670)
2.period			-0.0120 (0.0167)				0.00697 (0.0170)				-0.00174 (0.0235)	
3.period			0.00620 (0.0161)				-0.00852 (0.0165)				0.00875 (0.0227)	
4.period			0.0109				-0.0117				0.00524	

5.period			(0.0158)				(0.0161)			(0.0222)		
			0.0160				-0.0225			0.00338		
6.period			(0.0153)				(0.0156)			(0.0215)		
			0.000831				-0.0368**			-0.0231		
7.period			(0.0151)				(0.0154)			(0.0212)		
			-0.00120				-0.0321**			-0.0236		
8.period			(0.0149)				(0.0152)			(0.0210)		
			-0.0220				-0.0264*			-0.0455**		
9.period			(0.0149)				(0.0152)			(0.0210)		
			-0.0223				-0.0316**			-0.0519**		
10.period			(0.0145)				(0.0148)			(0.0204)		
			-0.0185				-0.0400***			-0.0612***		
11.period			(0.0145)				(0.0148)			(0.0204)		
			0.0417***				-0.0489***			-0.00497		
12.period			(0.0146)				(0.0149)			(0.0205)		
			0.0382**				-0.0386**			-0.00723		
13.period			(0.0149)				(0.0152)			(0.0209)		
			0.0342**				-0.0120			0.0263		
14.period			(0.0149)				(0.0152)			(0.0210)		
			0.0203				-0.00939			0.0148		
15.period			(0.0152)				(0.0155)			(0.0214)		
			-0.00337				-0.0165			-0.0177		
16o.period			(0.0155)				(0.0159)			(0.0219)		
			-				-			-		
Constant	0.118*** (0.0401)	0.0671 (0.0812)	0.161* (0.0901)	0.0904* (0.0533)	0.272*** (0.0448)	-0.115 (0.0817)	-0.150 (0.0920)	0.162*** (0.0591)	0.435*** (0.0573)	0.0650 (0.113)	0.0935 (0.127)	0.392*** (0.0754)
Obs.	663	663	663	663	663	663	663	663	663	663	663	663
R-squared	0.696	0.298	0.346		0.736	0.546	0.564		0.699	0.464	0.492	
N		43	43	43		43	43	43		43	43	43
Firm FE		YES	YES			YES	YES			YES	YES	
Year FE			YES				YES				YES	

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The dependent variable elicits capital structure measures such as: *stdr* is the short-term debt ratio calculated as short-term debt to total assets ratio; *lt dr* is the long-term debt ratio calculated as long-term debt to total assets ratio; and *tdr* is the total debt ratio calculated as total debt to total assets ratio The independent variables remain the same. Both the pooled ordinary least squares model and the fixed effects model are corrected for heteroscedasticity effects using both Breusch-Pagan and White methods, correspondingly. The figures as shown in the parentheses represent standard errors.

Reduced-form model of Ownership Structure

VAR.	<i>ln_ip_sh</i>				<i>ln_np_sh</i>				<i>ip_p</i>				<i>np_p</i>			
	OLS1	FE1	TFE1	RE1	OLS2	FE2	TFE2	RE2	OLS3	FE3	TFE3	RE3	OLS4	FE4	TFE4	RE4
mc_ev	0.437*** (0.113)	0.113* (0.0660)	0.147** (0.0673)	0.111 (0.0697)	0.190* (0.111)	0.0738 (0.0953)	0.147 (0.0957)	0.0895 (0.0998)	5.169*** (1.723)	0.653 (0.459)	0.561 (0.470)	0.657 (0.478)	-5.183*** (1.646)	-0.805 (0.678)	-0.445 (0.686)	-0.833 (0.704)
mcr	-0.805*** (0.181)	-0.672*** (0.225)	-0.641*** (0.227)	-0.847*** (0.221)	0.831*** (0.179)	-0.928*** (0.324)	-0.814** (0.323)	-0.151 (0.267)	-19.40*** (2.765)	-3.634** (1.561)	-3.841** (1.586)	-4.753*** (1.604)	12.98*** (2.641)	2.585 (2.305)	3.295 (2.316)	4.828** (2.308)
pti	0.0407*** (0.00358)	0.00543 (0.00361)	0.00659* (0.00366)	0.00944*** (0.00361)	0.0266*** (0.00355)	0.0118** (0.00522)	0.0156*** (0.00520)	0.0145*** (0.00455)	0.316*** (0.0549)	0.0194 (0.0251)	0.0158 (0.0255)	0.0361 (0.0259)	-0.329*** (0.0524)	-0.0377 (0.0371)	-0.0170 (0.0373)	-0.0755** (0.0374)
invtr	0.0286*** (0.00902)	0.00983* (0.00552)	0.00979* (0.00555)	0.00731 (0.00580)	-0.0530*** (0.00892)	-0.00958 (0.00796)	-0.00890 (0.00790)	-0.0221*** (0.00828)	0.610*** (0.138)	0.0454 (0.0383)	0.0423 (0.0388)	0.0504 (0.0399)	-0.692*** (0.132)	-0.130** (0.0566)	-0.127** (0.0566)	-0.142** (0.0587)
fa	-0.0108*** (0.00274)	0.0354*** (0.00855)	0.0268*** (0.00988)	-0.00764 (0.00587)	-0.0175*** (0.00271)	0.0658*** (0.0124)	0.0494*** (0.0141)	-0.0161*** (0.00511)	0.234*** (0.0419)	-0.370*** (0.0595)	-0.335*** (0.0691)	-0.304*** (0.0552)	-0.202*** (0.0400)	0.292*** (0.0878)	0.242** (0.101)	0.169** (0.0698)
indebt	0.00470** (0.00219)	0.00272** (0.00128)	0.00323** (0.00130)	0.00340** (0.00134)	0.00641*** (0.00217)	0.00178 (0.00185)	0.00319* (0.00185)	0.00382** (0.00192)	0.00956 (0.0336)	0.0233*** (0.00892)	0.0211** (0.00907)	0.0208** (0.00926)	0.0201 (0.0321)	-0.0181 (0.0132)	-0.0101 (0.0132)	-0.0128 (0.0136)
fatr	-0.000258 (0.00510)	0.0105*** (0.00305)	0.0115*** (0.00308)	0.00978*** (0.00322)	0.000988 (0.00505)	0.0123*** (0.00441)	0.0145*** (0.00439)	0.00727 (0.00459)	0.0952 (0.0782)	0.0292 (0.0212)	0.0288 (0.0215)	0.0316 (0.0221)	-0.122 (0.0746)	-0.0342 (0.0313)	-0.0260 (0.0315)	-0.0403 (0.0325)
mfer	1.137** (0.520)	0.268 (0.454)	0.356 (0.456)	0.370 (0.468)	0.337 (0.515)	0.254 (0.655)	0.482 (0.649)	0.689 (0.628)	12.16 (7.969)	-0.616 (3.154)	-0.850 (3.189)	-0.984 (3.272)	-3.448 (7.609)	0.0452 (4.656)	1.230 (4.656)	0.943 (4.782)
ccl	-0.0483 (0.0641)	-0.110*** (0.0411)	-0.106** (0.0417)	-0.0919** (0.0432)	-0.241*** (0.0634)	-0.193*** (0.0593)	-0.167*** (0.0593)	-0.150** (0.0614)	2.875*** (0.981)	0.0204 (0.286)	-0.0589 (0.291)	0.00178 (0.297)	-2.961*** (0.937)	-0.00589 (0.422)	0.206 (0.425)	0.0319 (0.437)
ctr	-0.0103 (0.00728)	0.00863 (0.00536)	0.00761 (0.00539)	0.00783 (0.00559)	0.0361*** (0.00720)	0.000132 (0.00774)	-0.00227 (0.00767)	0.00803 (0.00777)	-0.757*** (0.112)	0.0608 (0.0373)	0.0630* (0.0377)	0.0472 (0.0387)	0.908*** (0.106)	-0.0415 (0.0550)	-0.0526 (0.0550)	-0.00565 (0.0568)
roce	-0.128*** (0.0110)	-0.0191*** (0.00682)	-0.0172** (0.00685)	-0.0336*** (0.00706)	-0.0659*** (0.0109)	-0.0187* (0.00985)	-0.0148 (0.00975)	-0.0383*** (0.00999)	-1.152*** (0.168)	-0.0529 (0.0474)	-0.0583 (0.0479)	-0.0563 (0.0490)	0.999*** (0.160)	0.0621 (0.0700)	0.0839 (0.0699)	0.0770 (0.0717)
cf_rev	-0.0919** (0.0459)	-0.0356 (0.0243)	-0.0200 (0.0248)	-0.0499* (0.0255)	0.0379 (0.0454)	-0.108*** (0.0351)	-0.0925*** (0.0353)	-0.101*** (0.0368)	-1.780** (0.702)	0.237 (0.169)	0.237 (0.173)	0.248 (0.176)	2.081*** (0.671)	-0.229 (0.249)	-0.161 (0.253)	-0.240 (0.258)
qir	-0.00572 (0.0533)	-0.00932 (0.0263)	-0.0118 (0.0267)	-0.0140 (0.0280)	-0.0171 (0.0527)	0.0137 (0.0380)	-0.00419 (0.0379)	0.00870 (0.0411)	-0.344 (0.816)	-0.256 (0.183)	-0.215 (0.186)	-0.256 (0.191)	0.465 (0.780)	0.196 (0.270)	0.105 (0.272)	0.202 (0.282)
tang	0.765*** (0.239)	0.237 (0.183)	0.253 (0.186)	0.400** (0.189)	1.742*** (0.237)	0.409 (0.264)	0.341 (0.265)	0.848*** (0.257)	-8.535** (3.662)	-2.531** (1.271)	-2.226* (1.299)	-2.590** (1.319)	8.611** (3.497)	1.084 (1.877)	0.543 (1.897)	1.296 (1.928)
sz	0.894*** (0.0268)	0.154** (0.0625)	0.119* (0.0639)	0.494*** (0.0452)	0.695*** (0.0265)	0.276*** (0.0902)	0.267*** (0.0910)	0.769*** (0.0436)	5.225*** (0.410)	0.119 (0.434)	0.0413 (0.447)	-0.157 (0.408)	-4.216*** (0.392)	-0.228 (0.641)	-0.145 (0.652)	0.229 (0.525)
2.period			-0.0610 (0.101)				-0.105 (0.144)				0.565 (0.708)				-0.393 (1.034)	
3.period			-0.0879 (0.0980)				-0.168 (0.140)				0.470 (0.685)				-0.246 (1.000)	
4.period			-0.131				-0.289**				0.271				-1.493	

5.period			(0.0958)				(0.136)			(0.669)				(0.977)		
			-0.147				-0.353***			0.349				-1.568*		
6.period			(0.0931)				(0.132)			(0.650)				(0.949)		
			-0.195**				-0.406***			0.439				-1.907**		
7.period			(0.0917)				(0.130)			(0.641)				(0.935)		
			-0.201**				-0.444***			0.435				-2.223**		
8.period			(0.0905)				(0.129)			(0.632)				(0.923)		
			-0.161*				-0.435***			0.810				-2.548***		
9.period			(0.0905)				(0.129)			(0.632)				(0.923)		
			-0.0812				-0.389***			1.332**				-2.815***		
10.period			(0.0881)				(0.125)			(0.615)				(0.899)		
			-0.0571				-0.329***			0.876				-2.186**		
11.period			(0.0879)				(0.125)			(0.614)				(0.897)		
			0.0521				-0.195			0.745				-1.901**		
12.period			(0.0886)				(0.126)			(0.619)				(0.904)		
			0.0304				-0.228*			0.533				-1.550*		
13.period			(0.0903)				(0.129)			(0.631)				(0.921)		
			0.0640				-0.0921			0.0105				-0.679		
14.period			(0.0907)				(0.129)			(0.634)				(0.925)		
			0.0550				0.0418			-0.144				-0.335		
15.period			(0.0922)				(0.131)			(0.644)				(0.941)		
			0.0363				0.0421			0.107				-0.452		
16.period			(0.0944)				(0.134)			(0.660)				(0.963)		
			-				-			-				-		
Constant	9.119***	16.37***	17.03***	14.06***	9.978***	13.00***	13.68***	9.918***	6.075	81.44***	80.89***	82.16***	81.31***	22.83***	23.73***	21.76***
	(0.395)	(0.482)	(0.548)	(0.430)	(0.391)	(0.697)	(0.779)	(0.489)	(6.052)	(3.353)	(3.826)	(3.945)	(5.779)	(4.950)	(5.586)	(4.858)
Obs.	663	663	663	663	663	663	663	663	663	663	663	663	663	663	663	663
R-squared	0.745	0.250	0.269		0.699	0.331	0.367		0.467	0.244	0.255		0.441	0.090	0.123	
N		43	43	43		43	43	43		43	43	43		43	43	43
Firm FE		YES	YES			YES	YES			YES	YES			YES	YES	
Year FE			YES				YES				YES				YES	

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Put your notes here. The dependent variable elicits ownership structure measures such as: *ln_ip_sh* is the natural logarithm of the number of shares owned by Indian promoters (i.e. total stakes of individuals & hindu undivided family; central & state government promoters; corporate bodies; financial institutions & banks; and other indian promoters); *ln_np_sh* is the natural logarithm of the number of shares owned by non promoters (i.e. non-promoter institutions & non-promoter non-institutions); *ip_p* is the percentage of ownership stakes held by indian promoters; *np_p* is the percentage of ownership stakes held by non-promoters. The independent variables remain the same. Both the pooled ordinary least squares model and the fixed effects model are corrected for heteroscedasticity effects using both Breusch-Pagan and White methods, correspondingly. The figures as shown in the parentheses represent standard errors.

Capital Structure: Post Estimation Results

Description	Full model				Reduced-form model		
	stdr	ltdr	tdr		stdr	ltdr	tdr
Pooled OLS:				Pooled OLS:			
F (19, 643)	82.06	102.36	28.37	F (15, 647)	98.59	120.34	100.3
Prob > F	(0.0000)	(0.0000)	(0.0000)	Prob > F	(0.0000)	(0.0000)	(0.0000)
RMSE	0.0964	0.1069	0.1367	RMSE	0.0981	0.1098	0.1403
SSM	14.4873	22.2291	30.3382	SSM	14.2344	21.7737	21.6192
SSE	5.9746	7.3491	12.0185	SSE	6.2275	7.8045	12.7374
Fixed Effects:				Fixed Effects:			
F (19, 601)	15.22	41.43	28.93	F (15, 605)	17.13	48.45	34.97
Prob > F	(0.0000)	(0.0000)	(0.0000)	Prob > F	(0.0000)	(0.0000)	(0.0000)
sigma_u	0.1398	0.1987	0.1702	sigma_u	0.0873	0.1888	0.2120
sigma_e	0.0745	0.0747	0.1047	sigma_e	0.0757	0.0762	0.1057
rho	0.7787	0.8762	0.7254	rho	0.5703	0.8598	0.8008
Time FE:				Time FE:			
F (33, 587)	10.75	24.88	18.46	F (29, 591)	10.76	26.32	19.73
Prob > F	(0.0000)	(0.0000)	(0.0000)	Prob > F	(0.0000)	(0.0000)	(0.0000)
sigma_u	0.1369	0.1945	0.1701	sigma_u	0.0843	0.1915	0.2159
sigma_e	0.0725	0.0741	0.1027	sigma_e	0.0740	0.0756	0.1042
rho	0.7810	0.8732	0.7327	rho	0.5646	0.8653	0.8111
Effect test:				Effect test:			
F (14, 587)	3.49	1.61	2.7	F (14, 591)	3.06	1.73	2.28
Prob > F	(0.0000)	(0.0710)	(0.0007)	Prob > F	(0.0001)	(0.0454)	(0.0048)
Random Effects:				Random Effects:			
Wald chi2(19)	463.53	872.75	693.55	Wald chi2(15)	419.42	805.91	643.64
Prob > chi2	(0.0000)	(0.0000)	(0.0000)	Prob > chi2	(0.0000)	(0.0000)	(0.0000)
sigma_u	0.0535	0.0684	0.0807	sigma_u	0.0604	0.0788	0.0837
sigma_e	0.0745	0.0747	0.1047	sigma_e	0.0757	0.0762	0.1057
rho	0.3397	0.4563	0.3726	rho	0.3886	0.5168	0.3853
LM test:				LM test:			
chi2(1)	378.34	571.54	379.52	chi2(1)	467.98	766.00	505.36
Prob > chi2	(0.0000)	(0.0000)	(0.0000)	Prob > chi2	(0.0000)	(0.0000)	(0.0000)
Hausman test:				Hausman test:			
chi2(15)	60.4	74.13	72.43	chi2(15)	247.72	25.42	66.33
Prob>chi2	(0.0000)	(0.0000)	(0.0000)	Prob>chi2	(0.0008)	(0.0446)	(0.0000)

Ownership Structure: Post Estimation Results

Description	Full model					Reduced-form model			
	ln_ip_sh	ln_np_sh	ip_p	np_p		ln_ip_sh	ln_np_sh	ip_p	np_p
Pooled OLS:					Pooled OLS:				
F (18, 644)	109.05	85.36	32.04	28.37	F (15, 647)	125.69	100.07	37.84	34.05
Prob > F	(0.0000)	(0.0000)	(0.0000)	(0.0000)	Prob > F	(0.0000)	(0.0000)	(0.0000)	(0.0000)
RMSE	0.9538	0.9504	14.7860	14.1740	RMSE	0.9677	0.9576	14.8230	14.1550
SSM	1785.56	1387.81	126068.25	102575.52	SSM	1765.49	1376.28	124706.30	102326.76
SSE	585.81	581.69	140789.34	129379.01	SSE	605.88	593.22	142151.29	129627.77
Fixed Effects:					Fixed Effects:				
F (18, 602)	12.72	17.45	12.32	4.56	F (15, 605)	13.46	19.98	13.00	4.01
Prob > F	(0.0000)	(0.0000)	(0.0000)	(0.0000)	Prob > F	(0.0000)	(0.0000)	(0.0000)	(0.0000)
sigma_u	1.7792	2.0047	21.9330	19.9487	sigma_u	1.8889	2.0803	21.5102	19.3624
sigma_e	0.4436	0.6460	3.0838	4.5559	sigma_e	0.4502	0.6501	3.1293	4.6199
rho	0.9415	0.9059	0.9806	0.9504	rho	0.9463	0.9110	0.9793	0.9461
Time FE:					Time FE:				
F (32, 588)	7.82	11.53	7.25	3.35	F (29, 591)	7.48	11.8	6.96	2.86
Prob > F	(0.0000)	(0.0000)	(0.0000)	(0.0000)	Prob > F	(0.0000)	(0.0000)	(0.0000)	(0.0000)
sigma_u	1.7331	1.7566	21.6404	19.5784	sigma_u	1.8497	1.8810	21.3079	19.1991
sigma_e	0.4416	0.6321	3.0909	4.5188	sigma_e	0.4499	0.6401	3.1436	4.5897
rho	0.9390	0.8854	0.9800	0.9494	rho	0.9442	0.8962	0.9787	0.9459
Effect test					Effect test				
F (14, 588)	1.38	2.91	0.8	1.71	F (14, 591)	1.06	2.36	0.61	1.57
Prob > F	(0.1562)	(0.0003)	(0.6666)	(0.0501)	Prob > F	(0.394)	(0.0034)	(0.8611)	(0.0829)
Random Effects:					Random Effects:				
Wald chi2(18)	274.68	451.12	191.19	67.24	Wald chi2(15)	241.32	401.7	168.11	50.69
Prob > chi2	(0.0000)	(0.0000)	(0.0000)	(0.0000)	Prob > chi2	(0.0000)	(0.0000)	(0.0000)	(0.0000)
sigma_u	0.8192	0.4666	14.2431	12.7050	sigma_u	0.8299	0.5191	13.7468	12.1307
sigma_e	0.4436	0.6460	3.0838	4.5559	sigma_e	0.4502	0.6501	3.1293	4.6199
rho	0.7733	0.3429	0.9552	0.8861	rho	0.7727	0.3893	0.9507	0.8733
LM test:					LM test:				
chi2(1)	1665.95	559.02	2601.55	2288.01	chi2(1)	1806.7	639.96	2632.34	2299.68
Prob > chi2	(0.0000)	(0.0000)	(0.0000)	(0.0000)	Prob > chi2	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Hausman test:					Hausman test:				
chi2(15)	-175.75	203.6	-21.02	42.08	chi2(15)	64.39	127.43	-24.19	38.37
Prob>chi2		(0.0000)		(0.0000)	Prob>chi2	(0.0000)	(0.0000)		(0.0008)

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